

Learning Dexterous Manipulation: Hands, Simulators, Training, and Evaluation

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Abstract—A dexterous hand can change an object’s pose without setting it down. This survey covers 112 learning-based methods that attempt it, the hands they run on, the simulators they train in, and how they report results. The field has largely converged on one recipe: reinforcement learning on privileged state in a GPU simulator, distilled into a vision-only student for transfer to hardware. Almost half the corpus trains that way, nearly all of it in one simulator, whose own successors are barely used. A second and faster-growing branch learns from teleoperated demonstration instead, through vision-language-action models, diffusion policies, and systems whose contribution is the collection rig itself. Published work has tripled since 2022. Grasping remains the largest task family and bimanual coordination the fastest-rising, and almost every experiment runs on one of four hands. What the survey adds comes from reading released code and vendor specifications beside the papers. Eight hands that can be bought or built from published designs appear in no method paper here. No closed-loop policy in the corpus reports how far its hand passes into the object it holds, which a position-only success criterion cannot see. Section VII proposes a protocol for the unreported quantities; the body gives the counts behind every claim above.

Index Terms—Dexterous manipulation, multi-fingered hands, bimanual manipulation, reinforcement learning, imitation learning, physics simulation, contact modeling, benchmarking, reproducibility.

I. INTRODUCTION

A hand is dexterous when it can change an object’s pose without putting the object down. That property, and not the finger count, is what separates a hand from a gripper. Bicchi [1] draws the same line in Sec. 1.1 between restraining an object and manipulating it with the fingers: restraint is a static question about whether the contacts prevent motion, while in-hand manipulation is a dynamic one about whether contacts can be broken and remade while the object stays held. An et al. [2] put the same idea as the ability to reorient and manipulate objects through coordinated finger movement. A parallel-jaw gripper can hold almost anything and reorient almost nothing, and everything difficult about dexterity follows from that second question.

Analytic grasp theory settled the first question and never reached the second. Form closure has a first-order test on the grasp matrix and known contact counts, four in the plane and seven in three dimensions for any polyhedron [1, Sec. 1.3.1];

force closure adds the wrench balance and the hand Jacobian. Neither tells a controller what to do next, and the same chapter says why: force closure does not guarantee stability (Sec. 1.5), and “the nonsmooth nature of grasp dynamics, because of the unilateral constraints on displacements and forces, has made a thorough analysis very difficult” (Sec. 1.8). That tradition produced conditions to verify rather than policies to run.

Learned control did not solve those modeling problems. It substituted sampling a simulator for solving a model, and scoring a rollout for certifying a configuration. That trade is why the field moved, and it is also why this survey is organized as it is. Once a method stops proving properties and starts measuring outcomes, two things that used to come for free become the author’s responsibility: the simulator has to be faithful where fidelity matters, and the measurement has to mean something. Much of what follows asks how well this literature discharges those two obligations. One distinction runs through both and is worth naming early: a quantity measured on a reference trajectory is not the same quantity measured on a trained policy’s own rollouts, and this survey keeps that reference-versus-rollout split throughout.

One recipe now carries most of the field. Train a policy with reinforcement learning on privileged state, the object pose, velocities and physical parameters the robot cannot sense, in a GPU simulator running thousands of environments in parallel; distill that teacher into a student that sees only what a camera and joint encoders provide; transfer the student. Fifty-three of the 112 method papers in this corpus train with reinforcement learning, 35 of them in Isaac Gym, and 23 distill. In-hand cube reorientation is the recipe’s benchmark, inherited from [3] and still the task a new method is expected to show.

A second branch is growing underneath it and inverts the dependency. Instead of a simulator and a reward it takes human demonstrations through a teleoperation rig and fits a policy directly: 18 vision-language-action models, 14 diffusion policies, and 14 papers whose contribution is the collection apparatus itself. It is the newer literature, most of it from the last two years, and it trades the reward-design problem for a data-collection problem. Whether it displaces the simulator pipeline or ends up feeding it is the question this corpus cannot settle, but the growth is the most visible trend in it.

Both branches run on remarkably little hardware. Of the 33 hands with a documented specification here, four carry almost every experiment: the Allegro at 35 method papers, a Shadow or Adroit model at 21, the Inspire RH56 family at 19, and LEAP at 12. Eight further hands can be bought today or built from published designs and appear in the experiments

Correspondence: luai_abuelsamen@berkeley.edu. Every claim in this paper traces to a structured note taken from a parsed source, and every count is recomputed from those notes at build time. The corpus, the notes and the build that produce them are available from the author on request, and will be deposited with a persistent identifier.

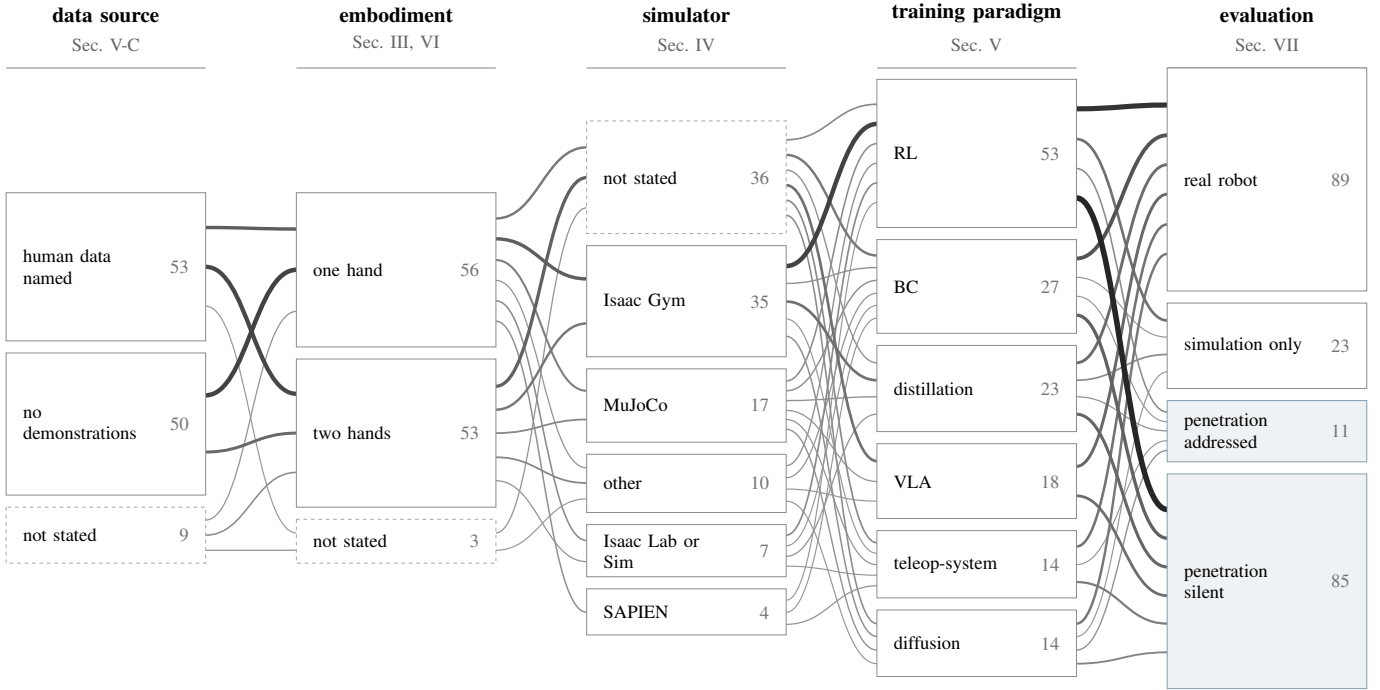


Fig. 1: The field on one page, and the map of this survey: nodes sized by corpus papers, edges drawn only for routes the corpus contains, each column head naming the section that covers it.

of none. That is not a manufacturing lag, since the hardware exists and is purchasable, and Section III takes it up: a field whose results rest on four platforms is more fragile than its publication volume suggests.

What this literature cannot currently do is compare its own results. Ninety-eight of the 112 methods state a success criterion, but not the same criterion; 70 state a trial count, so 42 do not; and the criteria run from never dropping the object to reaching a pose within a tolerance each paper picks for itself. Two success rates from two papers on nominally the same task are usually not measuring the same event, and none of the surveys preceding this one supplies a protocol that would make them comparable. Section VII proposes one, derived rather than asserted, with the trial counts a stated confidence interval actually requires.

This survey is built from a corpus rather than from memory. Two hundred and twenty-one bibliography entries, 218 of them read into a structured record under a fixed template, every value quoted from a parsed source with that source named, and every count printed here recomputed from those records at build time rather than typed in. Where a source is silent the record says so, which is why the coverage statistics in this paper are floors and not rates. The same discipline extended to released code: where a method published a repository, its reward was read from the repository as well as from the paper. Reading an implementation against its own description is established practice outside robotics, in repeatability and reproducibility studies [4], [5], in audits of how implementation detail moves reinforcement-learning results [6], in reward-design review [7], in benchmark re-releases [8], and most directly in recent work on paper-code alignment [9], [10]. We are aware of none that does it for a robotics field’s released reward functions, and

Section V-F reports what this one found.

The scope is single-hand and bimanual multi-fingered manipulation under learned control. Parallel-jaw work enters as a comparison and not as a subject. Two limits apply to every number here. This is a corpus of the learned era, which is a selection effect rather than a judgment on the analytic tradition; and a value this survey failed to extract is indistinguishable in the counts from a value its source never stated.

Fig. 1 puts the field on one page. What follows works outward from the task: the task families and what each demands (Section II), the hands and their sensing (Section III), the simulators and what they expose (Section IV), how policies are trained (Section V), what changes with two hands (Section VI), and how any of it is evaluated (Section VII). Section VIII states what the corpus supports and what it does not. Appendix A gives the method and the corpus, Appendix B the hardware and engine tables, Appendix C every paper-against-code comparison in full, Appendix D the coding of the fourteen surveys that precede this one, and Appendix E the derivation behind every count in the protocol.

II. A TAXONOMY OF THE PROBLEM

A. Task families

The six families below are read off the rows rather than imposed on them, and the last paragraph here says what they miss. Grasping is the largest. Fifty-seven of the 112 method rows carry the grasp label, the labels are not exclusive, and one paper can sit in several families. Success is a lift that survives a hold, and the thresholds differ by more than an order of magnitude. DexGraspVLA [11] requires the object “held 10 cm above the table for 20 s”, while Omnigrasp [12] requires it “held at least 0.5 s in simulation”. What makes grasping hard at scale

is the continuum of starting configurations. UniDexGrasp++ [13] states it in Sec. 4.3, that “we are dealing with an infinite number of tasks considering the initial object pose can change continuously”.

Functional and tool use is second with 45 rows. It is the family where a stable grasp can still be the wrong answer. Dexterous Functional Grasping [14] gives the case in Sec. 2.1, that “grabbing a hammer from the head or handle are both equally valid ways of using it”, and only one of the two lets the tool be used. Success is defined against the tool’s function, and the field has no shared way to state it.

In-hand reorientation has 31 rows and the most settled criteria, because the community inherited one number. OpenAI [3] declared the goal achieved below 0.4 rad of orientation error, DeXtreme [15] keeps 0.4 rad at test time against a 0.1 rad training tolerance, and Eureka [16] counts consecutive successes at 0.1 rad. A shared tolerance is not a shared protocol, because the stopping rule differs. Visual Dexterity [17] measures error “when the controller predicts it has reached the goal and stops”, which makes the policy a judge of its own trial.

Tracking a human reference has 12 rows and inverts the problem. The target is a trajectory rather than an endpoint, so success is a per-frame error band. ManipTrans [18] requires all four of 30 degrees of rotation, 3 cm of translation, 8 cm of joint error and 6 cm of fingertip error. The difficulty is that the reference came from a human hand and was never dynamically feasible for the robot. DexMachina [19] warns in App. B.4 that scoring by timesteps inside the thresholds makes “the results highly sensitive to the threshold values”.

Bimanual coordination has 43 rows and handover has 10. Handover is the smallest family and the one whose failure has a single moment, because the giver must release only after the receiver has the object. Dynamic Handover [20] reports a hit rate above its success rate and blames the gap in Sec. 5.4 on “occasional challenges encountered during the grasping phase of the catcher”. Not every paper here learns both sides. In [21] only the receiver is learned and the giver is a scripted arm.

Those six families do not cover the corpus. Twenty-four of the 112 method rows carry a label from outside them and eight carry no label from the six at all. Twenty fall in a catch-all class, mostly generalist policies evaluated on a task suite rather than on a dexterous task family, among them [22], [23], [24], [25] and [26]. Three adjacent families are named here rather than absorbed. Locomanipulation is [27], [28] and [29], where the base is not fixed and the gravity argument below changes character. Piano playing is [30], [31] and [32], discussed in Section VI, where success is a per-timestep F1 against a MIDI score rather than an object pose. Deformable manipulation is [33], whose object carries its own state and its own physics. Ferrari and Canny’s grasp-quality measure [34] carries no task label, because it is a measure rather than a task. Table I’s six rows are the families with enough papers to compare, and not a partition of the corpus.

B. Sources of difficulty

Contact is non-smooth, and that is a property of the problem rather than of any solver. Bicchi [1] states in Sec. 1.2 that

contact constraints are unilateral, and that losing a contact “involves an abrupt change of the structure of the model under consideration”. Its Sec. 1.4 gives the sharper version, that a rod sliding on rough ground has configurations with no consistent solution and configurations with more than one. Simulators inherit the difficulty. The re-implementation study [35] concludes in Sec. V that “there is no fully satisfactory approach at the moment, as all existing solutions compromise either accuracy, robustness, or efficiency”. Pang et al. [36] treat non-smoothness as the thing to remove, curating every contact pair in Sec. III-D so that contact points and normals change smoothly.

The hand has more actuators than the object has degrees of freedom and is still short of authority over it. The object moves only through contacts, and the contacts are what cannot be commanded. Bicchi [1] states in Sec. 1.4 that in multifingered grasps “the number of independent contact forces is much larger than the number of actuators. Thus, from a controllability standpoint, not all the contact forces are controllable”.

Vision is occluded by the hand doing the work. Bai et al. [37] name it in Sec. 4.3, that “occlusion hampers object tracking”. DexPoint [38] keeps vision and adds imagined hand points to the point cloud. Yin et al. [39] remove vision entirely and rotate objects from 16 binary touch sensors over the palm, links and fingertips.

Gravity direction changes the task rather than scaling it. Visual Dexterity [17] states that reorientation with the hand below the object “is much easier”, because “with a downward-facing hand, the hand must manipulate the object while simultaneously counteracting gravity”. AnyRotate [40] makes the direction a randomised variable by “randomly initializing hand orientations between episodes”, and its Sec. 5.3 measures the cost, with performance dropping progressively from palm up and palm down, through base up and base down, to thumb up and thumb down.

The second hand, and three counts. Three counts describe two hands and they measure different things. Fifty-three of the 112 method rows record two hands on the robot, which is all the corpus’s two-hand flag claims. Forty-three carry the task label for bimanual coordination, the narrower claim that coordinating the hands is the task. Section VI narrows again, to the 28 rows whose notes place a learned closed-loop controller on two multi-fingered hands, and its opening paragraph names every exclusion that takes the 53 down to the 28. That 28 is the denominator for every architecture count in this survey. Every bimanual claim here names which of the three it uses.

What the second hand adds. Four things genuinely change when the second hand arrives. Contact stays non-smooth, occlusion stays, and gravity stays the same problem.

Role asymmetry is one. AsymDex [41] assigns a dominant hand with full finger and wrist control and a facilitating hand with 6-DoF base pose only, so that “the facilitating hand repositions and reorients one object, while the dominant hand performs complex manipulations”.

Two hands on one object also close a kinematic chain through the object. In [1] Sec. 1.4 the hand and object dynamics are separate and are “linked through the n rigid-body contact constraints”. A second hand adds a second constraint set on

the same object rather than a second independent problem. That is where physical plausibility fails first. BimanGrasp [42] is the corpus row that measures it, rejecting a synthesised grasp when “total penetrations exceeded 1.5 mm” and reporting in Sec. IV-C that “penetration remains the primary cause of grasp failure”.

Two-arm collision does not exist for one hand at all. DyDexHandover [43] resets an episode when “any unintended arm contact with the environment or self-collisions” occurs. Bunny-VisionPro [44] pays for it in the controller, where adding a sphere-approximated self-collision cost raises motion-control time from 0.74 ms to 7.85 ms. Bidex [45] solves it in hardware, mounting the hands so the human arm and the teacher arm “are perpendicular to each other and do not collide”. DexMimicGen [46] states plainly that it does not handle inter-arm collision at all.

And the action space doubles. Bi-DexHands [47] runs 20 tasks on two Shadow Hands and finds single-agent PPO beating multi-agent RL on most of them, offering in Sec. 5.2 that “PPO algorithm is able to use all observations for training the policy, while MARL can only use partial observations”. AsymDex [41] attacks the dimensionality from the other side, halving the observation and action dimension through its role split and a frame relative to the facilitating hand’s object.

Two cells say the field has no agreed criterion, and both are in families where a second body is involved. A criterion naming a distance, a duration and a trial count can be re-run by someone else. A rubric cannot, and Section VII takes up what follows from that.

III. HANDS AND WHO MAKES THEM

A. Design axes

No degree-of-freedom, force, weight or price figure in Table IV or Table V, both set in Appendix B, was measured by anyone outside the maker. Six rows are the exception, from peer-reviewed papers with stated protocols: ILDA, Pisa/IIT, ORCA, RUKA, LEAP and BiDexHand. Every other claim is dated at its source. Table V prints the date beside each announced hand, and the repository tabulation carries a claim-date column for the hands that can be bought, running from Shadow’s specification of 4 December 2024 to vendor pages fetched on 17 September 2026.

The degree-of-freedom count is the first number a vendor states and the least comparable one. Shadow’s specification of 4 December 2024 reads “20 actuated DOF and a further 4 under-actuated movements for a total of 24 joints” [52]. Table IV prints joints first and actuated DoF second, and the gap is the informative number. Where a vendor’s own DoF figure differs from the joint count, the joint count is what the table prints, as with Inspire’s “Degrees of freedom 6, Numbers of joints 12” [53]. An actuator count is neither of those two, and the repository tabulation keeps it in a column of its own. DexHand states no joint count at all, only 16 finger micro-servos and two wrist servos, so 18 servos is the only count quotable for it [54]. Daxo states 120 actuators, no DoF figure, and a tendon structure with no rigid joints, which has far fewer kinematic degrees of freedom than actuators [55].

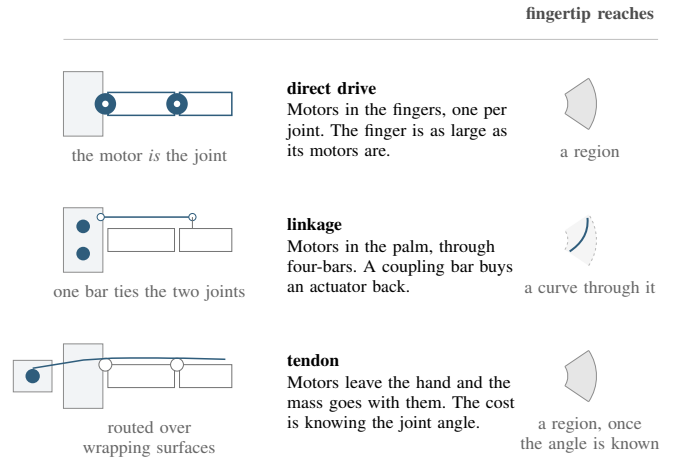


Fig. 2: Three answers to where the motor goes, and what each one costs. A coupling bar saves an actuator by tying two joints, so the fingertip reaches a curve through the reachable region rather than the region itself.

The actuation ratio is the real decision, and Fig. 2 sets the three answers side by side. The Pisa/IIT SoftHand drives 19 joints from one motor and its authors state the cost plainly: “No in-hand dexterous manipulation is required for this prototype” [56]. Tendon drive moves the motors off the fingers and the mass follows them. Shadow’s hand plus forearm weighs 4.3 kg, against 1.1 kg for the ILDA hand alone with every motor in its palm [52], [57]. ILDA states an 18 kg payload against Shadow’s 4 kg in a power grasp, and Shadow states no fingertip force at all. What tendons cost is state estimation. The Faive Hand estimated joint angles from tendon length through a Kalman filter and its cube reorientation failed on hardware, which its authors blamed on “poor joint angle measurement from the EKFs, especially when there is contact” [58]. Ruka-v2 adds detachable AS5600 encoders to measure that problem rather than close the loop on it, and those encoders put its open-loop linear joint-to-motor map at 8.26 degrees of error over seven joints [59].

Direct drive trades torque density for transparency. LEAP’s Dynamixel joints resist 19.5 N in a pull-out test against the Allegro Hand’s 8.5 N, at 595 g for the LEAP hand [60], and the Allegro’s own weight has no reachable source. Linkage drive is the argument against the forearm. ILDA puts 15 motors, drivers and ball screws inside the palm and reports 34 N at the fingertip in the bent pose and 28 N stretched [57]. Linkages buy coupling for free, as in BiDexHand’s four-bar that drives each DIP off its PIP, whose paper claims 16 actuated DoF and 21 joints while its README describes 15 servos driving 15 joints [61].

Fingertip force is the number a hand buyer reads first, and across these rows it is six different measurements. Table IV puts the quantity beside the figure: pull-out resistance for LEAP’s 19.5 N, pinch for RUKA’s 2.74 N and BrainCo’s 15 N, fingertip normal force under a 1 cm indenter for Unitree’s 10 N, a calibrated five-trial fingertip force for BiDexHand’s 2.14 N, a peak vendor claim for 1X’s 45 N, and two unlabelled spec columns for Sharpa’s “20 N | 12 N”. BrainCo’s whole-fist

TABLE I: Task families against the properties that define success.

task family	what is held	what moves	what gravity does	what counts as failure	shortest honest success criterion
In-hand reorientation	one rigid object, held by the fingers for the whole episode	the object's orientation in the palm frame, while contacts break and remake	a disturbance whose direction is set by hand orientation, and a load to counteract with the palm down, [17]	the object leaves the hand, e.g. the fall reset at goal distance 0.24 m in [15]	orientation error below a stated tolerance under a stated stopping rule, 0.4 rad in [3] and [15], 0.1 rad in [16]
Grasping	nothing at the start, the object rests on a support	the hand closes, then the object is lifted clear	the test itself, since the grasp exists to resist it	no lift, or a drop inside the hold window	a stated lift height held for a stated duration, 10 cm for 20 s in [11], 10 cm to episode end in [48], 0.5 s in [12]
Functional and tool use	a tool, held in the pose its function requires rather than the most stable one	the tool's own joint, or the tool acting on a second object	secondary, the function dominates	a stable grasp in the wrong place, [14]	no agreed criterion. Criteria range from never dropping the object, [14], to a four-point rubric, [49]. The one mechanical threshold is holding the tool joint past a commanded angle for the episode, [50]
Tracking a human reference	whatever the human held in the captured clip	hand and object follow a recorded trajectory under physics	full, and the reference never obeyed it, so the reference is not dynamically feasible	per-frame error leaves the band, or the object drops	all stated per-frame errors below threshold for the whole clip, 30 deg and 3 cm and 8 cm joint and 6 cm fingertip in [18]. Thresholds are not comparable across papers, and DexMachina [19] calls the results "highly sensitive to the threshold values"
Bimanual coordination	one object by two hands, or one object per hand	both hands, the object, and the relative pose of the hands	can be carried by one hand while the other acts, [41]	the object drops, the arms collide, or the hands work against each other	no agreed criterion. Task-specific thresholds dominate, e.g. 0.035 m block-to-cup distance in [41]. The one criterion with a physical-plausibility gate is 2.0 s of hold with total penetration under 1.5 mm, [42]
Handover and in-hand transfer	by the giver at the start, by the receiver at the end	the object crosses between hands, released only after it is regained	unopposed during the exchange, and total in a thrown transfer, [20]	a drop at the moment of release, and the catcher's grasp phase, [20] Sec. 5.4	the receiver holds the object and moves it clear, more than 10 cm from the first hand in [51], or holds it to episode end in [43]

grip of 50 N is a separate figure and is no longer in the same column as a pinch. ORCA's 19.6 N was none of these. It is the newton equivalent of a 2 kg index-finger payload at a fixed 600 mA motor current, so its force cell is empty and the figure is recorded as a payload instead [62]. Payload is no more comparable than force, which is why Table IV does not print it. Nine of its twenty hands state one, in the repository tabulation's payload column, and no two state it under the same test: Shadow 4 kg in a power grasp with no protocol given, Tesollo 2.5 kg pinching and 10 kg enveloping as rated figures, Unitree 3.5 kg palm-down and 4.5 kg palm-left on a 5 cm round object, ORCA 10.5 kg on four fingers and 2 kg on the index alone at a fixed motor current, BiDexHand 4.54 kg lifted, and RUKA 6.0 kg as the weight at which its joint-angle error passes 15 degrees.

Compliance is repairability seen twice over, and both are mechanical fuses: the Pisa/IIT SoftHand's rolling-contact joints return to assembly after over-extension and ORCA's "poppable" pin joints dislocate rather than break [56], [62]. ORCA also reports tendon drive as maintenance: "Prolonged use requires manual re-tensioning to maintain performance". What fails is rarely the motors. Its durability run found silicone skin degrading on two fingertips after about 2,000 to 4,000 grasp cycles and sensor wires snapping on three after about 4,500 to 7,000. The sensing wore out an order of magnitude sooner than the hand.

Two properties matter before any of the above, and neither is a column of Table IV: the repository tabulation prints a control rate and a URDF-or-MJCF column for every row, and this one does not. The first is the rate a loop can be closed at, which spans 250 Hz on a Tesollo to 1 kHz on a Shadow, a Unitree or a Wuji, with the Allegro at 333 Hz on its own CAN

clock and Inspire stating no rate at all. LEAP's 500 Hz is a serial query ceiling and its own sim-to-real policy runs at 20 Hz, which answer different questions [60]. A reader deploying torque control is bitten by that first. The second is whether a URDF or MJCF exists and who ships it, which is the bridge to Section IV and to this survey's recommendation to pick a hand a simulator already carries. Shadow, Sharpa, Wuji, the Ability Hand, RUKA, Faive and the Allegro driver ship one. LEAP is the warning: its paper says the URDF is released and the released API repository contains none [60].

B. Hands in use, and what they cost

The two oldest designs in Table IV carry 49 of the 103 method rows that name a hand at all, and seven rows use both. Neither design's date is confirmed by its own sources, so 2005 and 2016 are the bibliography's. Fig. 3 counts, per hand, the method papers whose own experiments use it. Of 112 method rows, 103 name a hand. The Allegro accounts for 35, Shadow for 21, the Inspire RH56 family for 19, a parallel-jaw gripper for 12 and LEAP for 12. Seventy-five of the 103 name an Allegro, a Shadow or Adroit model, LEAP or an Inspire. What no row states is how big any of them is. Neither hand table carries a dimension column, because no source behind them gives a hand's length or its width: the specification a reader could hold against their own hand is the one these sources omit, while the degree-of-freedom count, which all of them give, is compatible with a hand of any size at all. Mass is the nearest thing on record and it is not the same quantity.

The Allegro's position is the uncomfortable part. Its product page at allegrohand.com/v4 returned HTTP 404 while the Wonik wiki timed out [63]. Everything Table IV confirms about the most-used hand in the corpus comes from its ROS

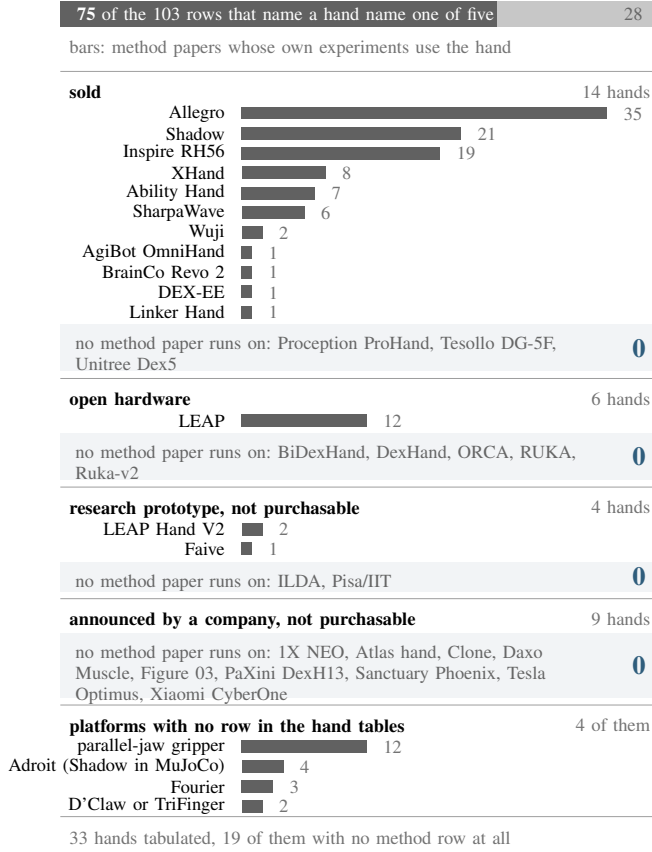


Fig. 3: Which hands the literature runs on, banded by how the hand can be got, with every tabulated hand shown and those no paper uses as named slots at zero. A paper naming two hands counts under both.

driver: 16 joints, four fingers, a torque interface, a 333 Hz CAN clock, 12 V, no tactile sensing. Its weight, joint torque, payload and price have no reachable source. The field’s most common platform cannot be specified from a page a reader can open.

Concentration would matter less if the hand did not move the result, and it does. On the same simulated cube rotation, LEAP reaches 0.2288 rad/s against the Allegro’s 0.0828 rad/s [60], and RUKA reports a 2.74 N pinch against the Allegro’s 1.60 N under the same three-trial pinch test [64]. A method compared only on Allegro hardware is compared at one point in a space where a single axis moves the headline number two or three times over. Inspire, XHand and Sharpa take 29 of the 103 rows between them, four rows use two of the three, and none is earlier than 2024. Eight take none at all: ORCA, RUKA, Ruka-v2, BiDexHand, DexHand, the Proception ProHand, the Tesollos DG-5F and the Unitree Dex5.

Open hardware and the collapse in cost. Six rows in Table IV are open hardware, and LEAP Hand V2 in Table V is a seventh. Five of the seven state a dollar cost: \$2,000 for LEAP, \$3,000 for LEAP Hand V2, \$1,500 for Ruka-v2, \$1,300 for RUKA and \$300 for DexHand. ORCA states a material cost below 2,000 CHF that the price column leaves unconverted, and BiDexHand states none [62], [61]. The Faive Hand states neither a cost nor a license that could be read, so it is not counted as open hardware here [58]. Two of the stated

bases need saying. DexHand’s \$300 is “additional total cost of components”, excluding the printing and the wrist servos [54], and ORCA’s own figure sits against Ruka-v2’s table listing ORCA at about \$3.5K [59].

The expensive end of the collapse is secondhand throughout. The only six-figure numbers on disk are RUKA’s comparison table at \$100,000 for a Shadow Hand and Faive’s “steep price tag of 110k GBP” [64], [58]. Shadow’s own page says to discuss pricing and Table IV’s price cell for it is empty. The collapse is real at the cheap end and secondhand at the expensive one, and it has barely moved the literature. Twelve of the 103 hand-naming method rows use an open-hardware hand, and all twelve are LEAP.

What the cheap hands give up is sensing. LEAP has none and names touch sensors as future work [60], RUKA states its design “lacks tactile sensing” [64], and BiDexHand’s sources never mention touch [61]. ORCA is the exception, with binary FSR fingertips whose threshold was measured as low as 0.05 N on a fresh fingertip, against the sensor’s rated 0.29 N, and 6.38 N on a degraded one [62].

C. Tactile sensing

DIGIT set the cost floor. Its paper states a “total estimated manufacturing cost is approximately 15 USD per sensor ... when manufactured in a batch of 1000” [65], which is where the price of touch sensing stopped being the obstacle. Durability was as much of the contribution: its gel degraded by 0.3 percent over fifteen abrasion passes, where the two gels compared against it degraded by more than 800.

Digit 360 is the same lineage at a different operating point, claiming about 8.3 million taxels, spatial features to 7 um, normal and shear force resolution of 1.01 mN and 1.27 mN, and on-device processing that cuts event-to-action latency from 6 ms to 1.2 ms [66]. It is also not for sale, and no corpus paper’s parsed text names it.

XELA’s uSkin is the non-camera alternative and the one that bolts onto hands the field already uses. Its curved fingertip kit for the Allegro V4 and LEAP carries 30 three-axis sensing points, a full Allegro integration reaches 368, resolution is 0.1 gram-force, and the kit runs at 275 Hz [67]. The 500 Hz figure often quoted belongs to the product family, not the curved fingertip. One corpus paper names it, [68], whose self-supervised encoders are pre-trained on 462.7k tactile images and beat matched end-to-end models by 95.1 percent when both see 33 to 50 percent of the labels. Its own bead-maze policies never complete the maze on the real robot.

The usage number is the one to keep, and it needs its inclusion rule stated: a sensor physically on the hand, read by the deployed policy. Eight of 112 method rows meet it: [40], [50], [69], [70], [51], [71], [72] and [39]. The rule excludes [73], whose 20 binary contacts are simulated on an Allegro without tactile hardware and whose released configuration disables tactile input; it also excludes [74] and [75] for the same reason. Sixty-three of the 112 method papers use the word tactile somewhere in their parsed text and 35 carry it in their structured note. Against the eight that meet the rule above, either number is the gap worth quoting.

Almost none of the eight uses a high-resolution sensor. Yin et al. [39] remove vision entirely and rotate objects from 16 binary touch sensors over the palm, links and fingertips, deployed zero-shot to a real Allegro, and Robot Synesthesia uses 16 force-sensing resistors read as binary [71]. Two papers exclude touch deliberately, HORA reporting rotation “even without the usage of vision and tactile sensing” [76], and ORCA’s authors dropping it from their reinforcement learning “due to the additional complexity involved in accurately modeling them” [62]. Taxel counts have risen by three orders of magnitude while the policies consuming them have stayed at binary contact.

D. The catalog against the literature

Three classes of evidence. Behind Table V’s rows sit three kinds of evidence, and conflating them is how a DoF figure with no source ends up in a survey. Vendor prose carrying numbers is the strongest, as on 1X’s page of 9 July 2026 [77]. Video is second and carries none. Press or bibliography assertion is third, as with Tesla’s V3 hand, known here only through a paraphrase of patents because the USPTO PDF parsed empty [78]. Table V blanks the cells resting on the third class and footnotes who did the arithmetic.

A survey can go one step past recording that a claim is unverified, which is to say which claims are implausible on their face. Daxo’s 120 actuators in 750 g is about 6 g per actuator including structure, tendons, routing and skin [55]. Clone’s 27 degrees of freedom under 2 pounds excludes a 500 W pump the source does not confirm is excluded [79]. Figure 03’s “Degrees of freedom, hands | 20” sits on the same tracker page as “Number of fingers | 10”, so it is almost certainly the pair [80]. Tesla’s repeated 22 is one article’s arithmetic of four DoF on each of five fingers plus two at the wrist, and Gen 2’s own figure was 11 [78].

Skepticism belongs to the evidence class, not to which table a row lands in. Sharpa’s 22 of 22, Wuji’s 20 of 20 and Tesollo’s 20 of 20 are vendor claims about unmeasured hardware and they sit in Table IV, where Sharpa’s page footnotes “Specifications may vary between products” and Wuji’s says the spec “will continue to iterate” on a Beta1 product [81], [82]. Even the best-specified vendor page in the corpus leaves its fingertip-force columns unlabelled.

What is sold against what is published on. The band at zero in Fig. 3 carries the finding. None of the nine company-announced hands in Table V appears in a single method row whose own experiments use it. Tesla, Figure, 1X, Sanctuary, Boston Dynamics, Xiaomi, Clone, Daxo and PaXini account for zero of the 112 method papers’ experiments. The nearest thing to a counterexample is [83], a Figure blog post claiming a 35-DoF whole-upper-body action space at 200 Hz that includes individual finger control. It never names the hand, gives no per-hand DoF count, and reports no success rate or trial count for any task. It is the maker describing its own unreleased hand, which is the evidence class the finding is about. The four research prototypes in Table V are in a different position, since the Faive Hand and LEAP Hand v2 Advanced account for three method rows between them [48], [45].

Table V’s emptiness is measurable and part of the same finding. Over the eight specification columns it prints, from

DoF to price, 58 percent of its cells are values no source stated, against 42 percent for the hands that can be bought; both captions state their own count. Over the twelve specification columns the repository tabulation carries, the same split is 61 to 39. The hands with the highest advertised DoF counts have the least specification behind them.

The one hand that crosses the gap crosses it because its maker published rather than announced. ByteDexter reaches a method row through a ByteDance technical report stating 21 DoF per hand, 16 piezoresistive fingertip channels in the action vector and real-robot success rates [84], not through a product page. That mechanism is available to every vendor in Table V and none has used it.

The gap runs the other way too, and the last column of Table IV shows it. Eight hands that can be bought or built from published designs take zero method rows each: the Unitree Dex5 with 94 pressure sensors on its P variant [85], the fully actuated Tesollo DG-5F [86], the 22-DoF Proception ProHand [87], ORCA, RUKA, Ruka-v2, BiDexHand and DexHand. Cheapness is established only for RUKA, Ruka-v2 and DexHand, because Table IV’s price cell is empty for Unitree, Tesollo, the ProHand, ORCA and BiDexHand, and the only Dex5 price on disk is Ruka-v2’s secondhand “~\$25K” [59]. A reader choosing a hand on this corpus’s evidence has four options with published precedent behind them: the Allegro at 35 method rows, a Shadow at 21 and 4 more for its Adroit model, the Inspire RH56 family at 19, and LEAP at 12, of which only LEAP has both a published design and a stated price. Below those four the record thins fast: eight rows for the XHand, seven for the Ability Hand, six for Sharpa, two each for Wuji and LEAP Hand v2, one each for five more, and none at all for the other 19 hands of Table IV and Table V. Those 19 sit in the tables on their specifications alone, and no method row in this corpus runs on any of them.

Model availability. Nineteen of the 33 hands in Table IV and Table V appear in no method row, and 11 of those are neither sold nor open and appear in none for that reason, so 33 is not the denominator for a software-lag claim: the eight hands of the paragraph above are. The rule that decides used from unused is one regular expression per hand against the method rows’ own hand field, in `tools/hand_usage.py`, so the partition can be recomputed rather than argued about. The lag runs through the tooling as well as the papers. Seven of the fifteen simulator rows ship a first-party hand model at all, six of those hands are real, Brax’s being a synthetic claw, and every one of the six is an Allegro or a Shadow; only four of the 33 tabulated hands appear in any engine. Bench2Dex [88] meanwhile compares 12 hands and DexVerse [89] six without stating a DoF count for any. A useful next step is a conformance suite for hand models: one URDF or MJCF per hand, with fixed joint-limit, mass and collision checks and a published pass or fail per engine.

IV. SIMULATORS AND THE PHYSICS UNDERNEATH

A. Requirements of a dexterous simulation

The clearest demonstration that hands are the hard case came from a benchmark that was not about hands. Erez et al. built a

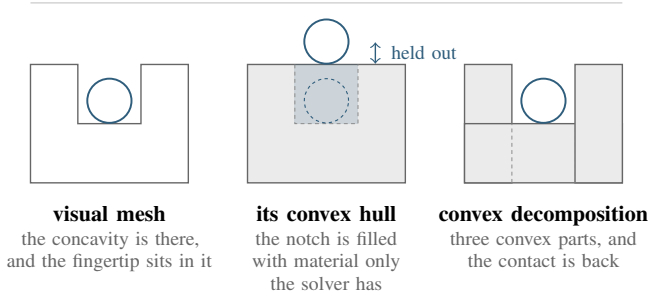


Fig. 4: The geometry the solver uses against the geometry the renderer draws. A single convex hull fills the concavity with material that exists only in the collision model, and holds the fingertip out of a contact the visual mesh allows.

35-DOF arm modeled on the Shadow Hand, closed it around a capsule with fixed spring-dampers, and asked four of the five engines for the largest timestep at which the object was still in the hand. MuJoCo held the grasp at 16 ms, PhysX at 2 ms, ODE at 0.25 ms and Bullet at 0.03 ms, a spread of a factor of 500 [90, Sec. IV-D]. Havok is the fifth and was excluded for want of a working PD controller. Three caveats travel with that spread. The engines run a deliberately restricted common model, hinge joints with sphere and capsule geometry, no boxes and no meshes (Sec. II). The authors call it “an open question whether this model system can be tuned to work better in the gaming engines” (Sec. IV-D). The timesteps are log-spaced and good only to a factor of two, and the authors wrote MuJoCo and disclose it.

A grasp is hard for nameable reasons. It is many contacts at once, all persistent, all near stiction, on an object much lighter than the mechanism holding it. Persistence and multiplicity make the problem hyperstatic, and Le Lidec et al. show that per-contact solvers of the projected Gauss-Seidel family, and RaiSim’s, then inject spurious jamming forces at stiction that vanish only once the object slides. The mass ratio makes it ill-conditioned, and in their stacked-cube test at a 10^3 to 10^{-3} kg ratio those same methods fail to converge. Global methods with proximal regularisation stay robust in both cases [35, Sec. IV-A]. What degrades a truncated solve is therefore conditioning and redundancy, not stiffness, and a grasp supplies both by construction. Locomotion is forgiving by comparison. On flat ground the contact model and solver choice “hardly affects” their quadruped’s tracked base velocity, and only on rough, slippery terrain do RaiSim and CCP deviate. A walking robot makes and breaks a few contacts against ground far heavier than itself. A hand does neither.

Speed enters by the same door. MuJoCo Playground reports that contact time scales with the number of possible contacts rather than the active ones, because JAX requires static shapes, which is why its tasks override the bound on contact points and the bound on geometry pairs by hand [91, Sec. VI]. Every override printed in that paper is a locomotion port. A hand’s bound would have to be set the same way, for the worst case, but no hand configuration is printed.

Fig. 5 sets out the stages of one simulation step. Three of them make overlap, and they do not answer to the same knob. In an engine that enforces non-penetration at the velocity level,

integration turns any residual approach velocity into overlap of order v times Δt ; an engine that enforces the gap at the next configuration carries no such term, which is why Dojo’s hard-contact NCP keeps its feet above the floor at every timestep it was tested at. Constraint assembly fixes the compliance a loaded contact then rests at. A truncated solver leaves a residual that grows with conditioning. Which of the three dominates in a grasp is not measured anywhere in this corpus, and the three are not ordered here. A fourth thing is not on the figure, because it is not a stage of the step and not a source of overlap: a mismatch between the geometry the solver uses and the geometry the renderer draws, which runs in both directions and is drawn in Fig. 4.

B. Contact models and solvers

Table VI, set in Appendix B, is the engine-by-engine comparison, one row per simulator, built only from what a note confirmed. Le Lidec et al. supply the taxonomy that organises it, checking each formulation against the Signorini condition, Coulomb’s law, and the maximum dissipation principle. Linear complementarity, the family of Bullet, ODE, PhysX and DART, satisfies Signorini alone, because linearising the friction cone to a pyramid biases friction toward its corners. The cone complementarity problem satisfies the other two but relaxes Signorini, so contact acts at a distance of size $\Delta t \cdot \mu \cdot \|c_T\|$. The full nonlinear problem satisfies all three and is non-convex, by Table II of [35]. A contact model is not the algorithm that solves it. PhysX linearises the cone, which places its model in the LCP family, and it does not solve an LCP. Its Temporal Gauss-Seidel scheme folds substepping into the Gauss-Seidel sweep and exposes position and velocity iteration counts separately [92, Sec. 3].

MuJoCo is inside that taxonomy rather than outside it. Le Lidec et al. classify it as CCP-MuJoCo, a convex relaxation solved by a Newton method on the primal QCQP, in the same family as CCP-Drake, the SAP-style scheme, by Table III of [35]. The relaxation makes two errors of opposite sign in one engine. A loaded contact carries a violation, and a sliding contact carries force at a positive gap. Drake’s SAP inherits the same pair, and its gliding effect at distance $\varphi \approx \delta t \cdot \mu \cdot \|v_t\|$ “unfortunately does not go away as $\delta t \rightarrow 0$ ” [93, Sec. V-A]. Le Lidec et al. call that compliance a “numerical trick designed to circumvent the issues due to hyper-staticity or ill-conditioning at the cost of impairing the simulation”.

The depth a loaded contact carries is a chosen number, not a property of the engine. In a regularised formulation the steady-state violation at a contact is the normal load times the compliance: zero at zero load, growing with what the contact carries. MuJoCo drives that violation with a critically damped stabiliser whose steady-state depth, for an object resting under gravity, has a closed form independent of the object’s mass [94, Sec. V]. Whoever sets the stiffness sets the depth.

Published penetration figures do not compare with each other, and the reasons are instructive. Dojo solves a hard-contact complementarity problem and stays above the floor at every timestep it was tested at, where MuJoCo penetrates tens of millimetres on the same Atlas drop [95, Sec. V-A].

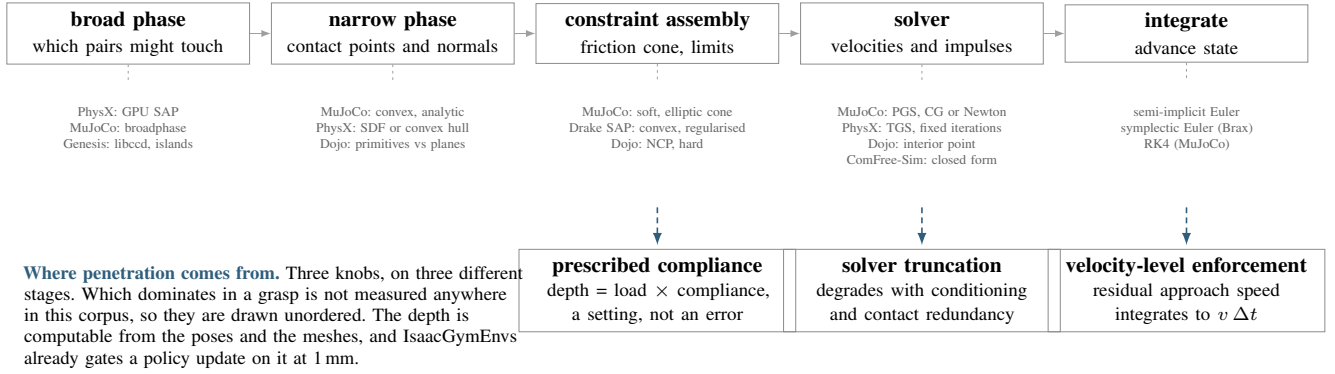


Fig. 5: One simulation step, and where engines differ: the stages every rigid-body engine runs, with the choices that separate the engines used for hands under each; dashed arrows mark the three stages that overlap.

But the MuJoCo column is not a trend: a ten-times-smaller step produces more overlap, which no timestep-independent stabiliser does, so either that configuration ties compliance to the timestep or the quantity is an impact transient rather than a steady-state depth. Drake’s SAP reports figures orders of magnitude smaller again, and they are analytical bounds for a point mass at rest on a plane under a near-rigid stiffness rule, not measured depths [93, Sec. V-B]. ComFree-Sim does report penetration directly, as an explicit tuning parameter, on 5 cm convex primitives dropped under their own weight [96, Sec. IV-A]. A point-mass bound, a humanoid drop transient and a dropped primitive differ in load, inertia and regime, so the distances between them measure nothing. What the set does establish is the part that matters for a hand: in a compliant formulation the depth follows from a stiffness somebody chose, and the GPU era moved that knob rather than removing it. [90] puts the consequence plainly, reporting that engines pushed past their stable step “effectively simulate a different physics model which can no longer hold the object” (Sec. IV-D). No paper in this corpus reports the depth for a fingertip loaded by a grasp, which is the one configuration a dexterous result depends on.

That brings Table VI’s most important column, and the claim it does not support. The column records whether a parsed source reported a penetration depth, not what an engine can compute. Two of the 15 engines report one, Dojo and ComFree-Sim, and both are engines whose paper is about contact accuracy. Four are recorded as not reporting it: Brax, Isaac Gym, Isaac Lab and Orbit. Nine rows are blank. The word “penetration” appears nowhere in the Isaac Gym paper, and Isaac Lab’s contact sensor reports force, duration and an average contact point with no contact-quality metric. The Isaac family carries 42 of the 112 method papers in this corpus.

The tooling exists and the number is not recorded. NVIDIA’s own IsaacGymEnvs computes interpenetration depth in simulation: its IndustReal tasks sample points on one mesh, query them against the other, and reduce to a per-environment maximum, then gate the policy update on it, scaling reward down as the maximum approaches a one-millimetre threshold. That is a shipped task measuring penetration per environment during training and acting on it, in the engine Table VI records as not exposing the quantity. The depth is computable from the

poses and the meshes in a few lines of Warp, and the field’s own benchmark repository already does it.

One knob in that stack bears on penetration and nothing in the corpus explains its setting. PhysX’s separation-velocity cap governs how fast the solver pushes overlapping bodies apart, so it sets how long an overlap persists rather than how deep it gets, and it appears across unrelated stock IsaacGymEnvs tasks at five different values [97]. What the corpus shows is the pattern of values; why any of them was chosen is not in the record.

The rest of Table VI is largely empty, and the emptiness is a result. Forty of the 105 cells in its seven property columns, 38 percent, are values no parsed source stated, which is what its caption counts. The repository tabulation is emptier still, because it carries four columns this one does not: 69 of the 150 cells outside its engine column, 46 percent. No engine paper states a default physics timestep. Three report one for a named experiment, and the timestep column reports those experiment settings. Isaac Gym’s cell is its Shadow Hand step, from the only per-task timestep table any engine paper here publishes, which runs 1/120 s for Shadow Hand and Allegro, 1/200 s for ANYmal and TriFinger and 1/60 s for Franka [92, Table 4]. MuJoCo’s 0.01 s is the 27-DoF humanoid test’s step and ComFree-Sim’s 0.002 s is its benchmark step, against a stated stability limit near 0.02 s. Seven of the fifteen engines ship any dexterous hand at all. Two of the columns only the repository tabulation carries are worth the trip: four engines state a solver iteration count there, and two state a license, which is the question a reader choosing an engine actually has.

C. GPU-parallel engines and their throughput

The GPU-parallel turn. Isaac Gym set the pattern. Physics, observations, rewards and actions stay on the GPU, and PhysX resolves contacts with the Temporal Gauss-Seidel sweep described above. Its per-task timesteps are published, which is rare: the Shadow Hand runs a 1/120 s physics step under a 1/60 s control step, or 1/20 s in the OpenAI variant. The result that reorganised the field is that reproducing OpenAI’s Shadow Hand cube reorientation took under an hour on one A100, against 30 hours on 6144 CPU cores and 8 V100s [92, Sec. 6.4.1]. Thirty-five of the 112 method papers in this corpus run on it.

Orbit and Isaac Lab moved the stack to PhysX 5, and the dexterous offering is thinner than the predecessor’s: the first-party suite is lifting, grasping and reorienting with the KUKA Allegro hand, while Shadow and Ability hands appear only through third-party grasp evaluation [98, Sec. 7.2.3]. MJX and MuJoCo Playground took the other route, putting MuJoCo’s own solver on the GPU through JAX at the cost described above. MuJoCo Warp ports the same solref and solimp soft-constraint rows to NVIDIA Warp, and its README states that the legacy PGS solver and the noslip pass are unsupported and that differentiability “is not yet available” [99]. GPU MuJoCo is not a differentiable MuJoCo. Genesis is the broadest, pairing a rigid constraint solver with FEM, MPM, PBD and SPH solvers and a differentiable backward path, and it ships a Shadow Hand. Its README quotes no throughput number at all, so the large FPS figures that circulate for Genesis trace to nothing in [100].

Newton changes how these names should be read. It is explicitly multi-solver: SolverMuJoCo wraps MuJoCo Warp, SolverKamino is a proximal-ADMM and dual-variational-inequality contact solver, and XPBD and VBD do position-based constraint projection [101]. Its contact model is solver-dependent, not a property of the engine, which is how Table VI records it. Isaac Lab already offers a Newton–MuJoCo Warp backend in its hands demo, and its roadmap announces Newton integration. An experiment reported as Isaac Lab may be running PhysX 5 with TGS, or MuJoCo’s soft constraint rows under Warp, and those two make different contact errors. The name also fails to fix the physics inside one engine. MuJoCo’s convex solver is a family, interior point or projected Newton, conjugate gradient or Gauss-Seidel [102, Sec. II-D], and MuJoCo Warp supports neither PGS nor the noslip pass. PhysX 4 and PhysX 5 differ in whether non-convex rigid bodies get SDF collision, which is a contact-geometry difference between Isaac Gym and Isaac Lab under one vendor name [103], [98]. Naming a simulator no longer names its physics. Papers should report the backend and the solver beside the framework.

Reported throughput, and why the numbers do not compare. Four headline figures measure four different quantities. Isaac Gym reports parallel environment steps per second with physics, observations and rewards on device: 150,000 for the Shadow Hand at 16,384 environments on one A100. Isaac Lab reports frames per second in training, which includes the learning update: over 900,000 for the DextrAH teacher task at 16,384 environments on eight RTX Pro 6000 GPUs, with no single-GPU dexterous number anywhere in the paper. ManiSkill3 reports “up to 30,000+ FPS” with RGBD and segmentation on one RTX 4090, measured over 1000 random actions with reward and termination removed from the timing, and does not state the environment count behind the figure [104]. Orbit reports a 125,000 FPS physics-only ceiling on an RTX 3090, also with no environment count [103, Sec. VII].

The one number reported cleanly enough to reuse is MuJoCo Playground’s, in PPO steps per second on a single A100 over five seeds. LeapCubeReorient runs at $76,354 \pm 143$ and PandaRobotiqPushCube at $487,341 \pm 4,346$. Same hardware, same measurement, same codebase, a factor of 6.4 between

two environments. It is not a factor from the task in isolation. Playground tunes solver iterations, line-search iterations, timestep and contact bounds per environment, with values as far apart as one and four solver iterations, by Table III of [91], and it does not print the two configurations side by side. That omission is what makes the two figures incomparable. The same confound sits inside Isaac Gym’s paper on one A100: 700,000 environment steps per second for Ant, 200,000 for Humanoid, 150,000 for the Shadow Hand. A training-loop FPS is also mostly not a measurement of physics. Playground breaks the fractional cost down on an RTX 4090: for CartpoleBalance, physics is 0.02, rendering 0.06, inference 0.01 and the policy update 0.91, and the policy update still dominates on the Franka task.

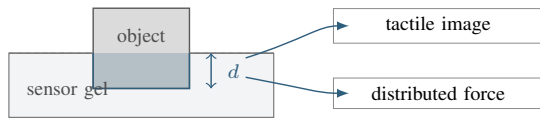
Cross-framework comparisons add further free parameters. ManiSkill2’s PickCube table takes the best result over 16 to 512 environments for each system [105, Table 1a], giving ManiSkill2 with a render server 2487 ± 24 FPS at its optimum of 64 environments against Isaac Gym’s 865 ± 35 at its optimum of 512. The env-count tuning is the second problem here. The first is that the two systems are not the same kind of thing: ManiSkill2 runs rigid-body physics on CPU worker processes behind a shared GPU render server, Isaac Gym runs physics on the GPU, and its own paper says so plainly. The measured quantity includes 128×128 rendering at 500 Hz simulation and 20 Hz control for both, so this is a visual sample-collection loop rather than two physics engines. Its authors add the fidelity caveat themselves, and Playground is equally explicit that its cross-simulator plot borrows its Isaac Lab and ManiSkill3 numbers from the ManiSkill3 paper. Several sources give no number at all: MuJoCo Warp’s README points to an external nightly dashboard, and Newton’s and Genesis’s READMEs contain no FPS or speedup anywhere.

The failure reaches past the engine papers into the methods. Of the 47 corpus method papers that name a GPU-batched simulator, 19 state no environment count anywhere, and Isaac Lab’s own paper states no physics timestep, no decimation and no solver iteration count for any task. A throughput figure is interpretable only with the environment count, the GPU, the timesteps, the solver iteration budget, and a statement of whether rendering and the learning update sit inside the measurement. Almost nobody reports all five.

D. Tactile simulation and the sim-to-real gap

The three tactile simulators in this corpus calibrate against three different things, and none of them is a manipulation outcome. TACTO is a rendering layer over a host engine, by default PyBullet’s rigid contact model. It reads post-solve link poses and the engine’s reported normal force and maps that force to gel-mesh deformation at the rendering level, so it contributes no contact physics of its own. Its only sim-to-real number is a tactile pose-estimation task, at 1.66 ± 0.16 mm with color-jitter augmentation against 0.76 ± 0.07 mm for a model trained on 128 real datapoints, by Table II of [106].

Taxim is example-based rather than simulated, with an optical model calibrated from 50 real indentations, and it beats TACTO on every optical-similarity metric against real images.



d is the depth by which the two solids overlap: the quantity every other engine in Sec. IV works to remove

the same d , thresholded, is the contact flag:
on above 0.5 mm, off below 0.2 mm



The inversion. Everywhere else in this survey the overlap is the error. For a compliant tactile sensor it is the signal, so the quantity a policy elsewhere is penalised for producing is the one this sensor is calibrated to read.

Fig. 6: A compliant tactile sensor reads the overlap depth. The tactile image, the distributed force and the binary contact flag are all functions of the one quantity every other engine in this section works to drive to zero.

Its marker-motion model is calibrated against an ANSYS FEM, and the errors are 1.00×10^{-2} mm real against FEM, 1.02×10^{-2} mm real against Taxim, and 3.96×10^{-3} mm FEM against Taxim. Taxim tracks its own reference more tightly than that reference tracks reality, so its residual is bounded below by the calibration target. The authors also state that they simulate quasi-static contact only, with slip left as future work [107, Sec. V]. Slip is what a hand needs.

Tactile Genesis inverts the thread. It implements two penetration-depth backends on the collision geometry, an analytic SDF query and a BVH raycast, and uses the depth as the sensor signal rather than as an error, so the quantity every other engine treats as a defect sets this one’s operating point.

The sim-to-real gap for hands. Most of what is written about this gap is attribution without measurement. PenSpin asserts that the pure physics gap “cannot be bridged by extensive domain randomization alone” while reporting no experiment that isolates it [73]. MuJoCo Playground attributes its LEAP hand failures to physical flex in low-cost hardware and names more accurate collision geometries as the fix, without measuring either. DeXtreme lists four candidate causes for its shortfall, including a malfunctioning Allegro thumb used in most trials, and disambiguates none by experiment [15].

What has been measured is narrower, and Section VII collects the transfer numbers. The one result that links a quantified model error to transfer is a humanoid recipe paper, which ranks its system identification runs by dynamics-model MSE and pairs each with real success over 10 trials: the lowest-MSE model grasps 8 out of 10, the median-MSE model 3 out of 10, the highest-MSE model 0 out of 10 [108, Table 1]. Where a cause has been pinned down elsewhere it is usually perception or actuation, not contact. OpenAI’s pose estimator has 3.12 mm error on rendered images and 9.27 ± 4.02 mm on 992 real ones, and PDDM reports a camera tracker with 5 mm average error and 20 ms latency as the unmodelled source in its real numbers [109, App. C].

The one paper that looks at the contact side is usually

read backwards. DeXtreme’s real-to-sim replay interpenetrated because the replayed poses carried the pose estimator’s error, not because the contact model failed, and the paper says so: it could not calibrate physics against the real system because the state it replayed was already wrong. The commonly cited reading, that the contact model was at fault, is not what the sentence says.

V. TRAINING A DEXTEROUS POLICY

A. Sources of supervision

A dexterous policy is defined by what supervises it, and four sources are in use across the corpus’s 112 method rows: a reward function supervises reinforcement learning, a human demonstration supervises imitation, a human reference trajectory supervises a physics-based tracker, which is imitation with a simulator in the loop, and nothing supervises a model-based planner, which is handed a cost and a model instead.

The labels do not partition the corpus. Reinforcement learning is the largest at 53 of the 112 rows, behavior cloning next at 27, distillation at 23, and ten further tags cover the rest. They sum to far more than 112, because most methods published since 2024 sit on two branches at once: a reinforcement-learning teacher and a behavior-cloned student are one paper, not two. Fig. 7 draws the tree and the cross-links. The repository carries the row-by-row version of the same thing, one row per method, and that is what to scan when looking for work comparable to your own.

What the deployed policy looks like once training is done matters more than the name a paper gives itself, and on that axis the field has converged hard. Section V-E shows why.

What the tabulation does not record. The emptiest columns of the method tabulation are the ones a reader most needs: only 31 rows state an environment count, only 70 state how many real trials are behind the headline number, and only 39 state how many unseen objects were tested. The trial and unseen-object figures are the audited ones, after Section VII-A recovered 15 trial counts and 7 unseen-object counts that the notes carried and the extraction had dropped. A mostly empty row is not a weak method, but it is one that cannot be compared with any other row here. All 112 rows carry the columns this survey reads, and Appendix A says where they are and how to read them.

B. Reinforcement learning

The standard recipe. Fifty-nine of the 112 method rows learn from a reward. Forty-eight name PPO as the algorithm, and the next most frequent, DAPG, appears four times. Forty-three run their own experiments in a GPU-parallel simulator from the Isaac family or Genesis, and 36 do both. Only 31 rows state an environment count, with a median of 8192 and a maximum of 64000. Twenty-three rows distill a privileged teacher into a deployable student, and 16 combine all three of PPO, a GPU simulator and distillation. That 16-row intersection is the recipe as it is actually practised.

The hand is almost always an Allegro. It appears in 35 of the 112 method rows, and its hold on in-hand reorientation



Eight further rows carry a trajectory-optimisation or MPC tag inside a learned pipeline and are counted on the branch that learns. The double-dashed edge is the one overlap that is a theorem rather than a pipeline: pang_global_planning_2022 proves that a policy gradient and an analytic log-barrier relaxation compute the same local model of contact.

Fig. 7: What supervises a dexterous policy, subdivided to where the 112 method papers actually differ; each leaf names two of its corpus keys and counts the rest, bars scaled by row count, dashed curves marking overlap.

is tighter still: more than half of those rows use it, against roughly a fifth for the Shadow and an eighth for LEAP.

Privilege is handled two ways. OpenAI [3] gives the value network object and target orientation, joint angles, joint velocities and object velocities the policy never sees, with a footnote recording that current object orientation was left out of the policy inputs by accident, and DeXtreme [15] uses the same asymmetric critic with a 2048-unit LSTM against the actor’s 1024. Hora [76] instead compresses nine privileged object properties into an eight-dimensional vector regressed from 30 steps of proprioception, and Visual Dexterity [17] distills twice, from a state teacher to a synthetic point cloud and then to a rendered one, for a fivefold speedup.

Domain randomisation is the part of the recipe with the least discipline. OpenAI [110] made it adaptive, pushing each range boundary out when performance at that boundary exceeds 20 successes and pulling it in below 10, and DeXtreme [15]

reproduced the mechanism with a 256-sample queue and 40 percent of environments dedicated to boundary evaluation. Both publish the discovered ranges. Below that standard, two shipped configurations disagree with their own papers about what was randomised: Hora [76] states a joint-noise range of $U(0, 0.005)$, while its public configuration uses 0.02; its README warns that this revision does not reproduce the paper. PenSpin [73] describes random disturbance forces, while the only public configuration that names them disables them. DexPBT [111] is consistent: its configuration disables randomisation, as the paper says.

Reward engineering. The repository puts the 21 in-hand reorientation methods against nine recurring term families and marks each cell by where the term was found: in the paper, in the code, in both, or in the code with every shipped configuration setting its weight to zero.

The families are not equally popular. Nineteen of 21 methods

have a goal or rotation tracking term, which is the task. Thirteen penalize effort as torque, work or joint velocity, 13 pay a sparse success bonus, 11 penalize object velocity and 10 penalize a drop. Eight penalize deviation of the hand from a canonical grasp pose, a family the plan for that matrix did not anticipate and which had to be added. Six penalize action rate or magnitude, five reward closing the distance from fingertips to the object, and four carry a contact or force mark at all.

Three marks an earlier draft recorded as *code* now read *code (0)*, because in each the term is in the released code with every shipped configuration setting its weight to zero: DeXtreme [15]’s timeout reward, DexPBT [111]’s fall penalty, and PenSpin [73]’s action penalty. Marking them *code* would tell a reader the code optimises something the paper does not state, and leaving them unmarked would hide a term that is in the file. The repository carries the config key and the zero beside each of the three marks.

Only three of the four contact marks are the finding. AnyRotate [40] scores good and bad fingertip contacts, Lin et al. [112] reward a friction-cone wrench margin, and Zhao et al. [113] track a commanded grasp force. The fourth, [17], penalizes the *object* touching the table, a task-shaping term against using the table as a third finger rather than a hand-object term, and its table-contact flag defaults to off, with no shipped config setting it true. Three of 21 in-hand reorientation methods, then, put a hand-object contact or force quantity in the reward, and none puts interpenetration in it. Li et al. [114] penalize interpenetration with a differentiable signed-distance term, but during offline reference construction, not in the policy’s reward. Across all 112 method rows, 85 do not address penetration at all, 5 constrain it, 3 measure it, 3 penalize it and 16 say nothing either way. The physical quality of the contact is not something this literature optimises.

Term counts range from one to ten. DemoStart [115] gives a single binary terminal reward of 1 and lets a curriculum over demonstration states do the shaping. AnyRotate [40] and ViserDex [116] each publish ten weighted terms. ViserDex is the cleanest specification of the 21, with every term, weight and equation in one appendix table, and a statement that the same weights are used for every object.

Weights are frequently unrecoverable. Three of the 21 rows give no numeric weight that survives PDF conversion, because the equation blocks are images, which is a limit of this survey and is recorded as such. Robot Synesthesia [71] is a worse case. Its six coefficients are described only as “tuned hyperparameters” and no number is printed anywhere in the paper, so the reward it reports cannot be reconstructed from the paper. DexReMoE [117] names an angular-velocity penalty weight in prose with no entry in its hyperparameter table, and DexTrack [97] names an affinity reward that is missing from its weight table.

Curricula, populations and machine-written rewards. Curricula in this literature relax physics or tighten tolerances. DexPBT [111] tightens a success tolerance from 0.075 to 0.01 by a factor of 0.9 every 3000 environment steps once three successes are logged, and ManipTrans [18] starts at zero gravity and high friction and restores both while narrowing a fingertip threshold from 6 cm to 4 cm. DexMachina [19]

attaches virtual spring-damper controllers to the object, lets them do the task at first, then decays their gains to zero. The two that ablate the curriculum report it is load-bearing: RotateIt [72] sets its rotation-penalty weight to zero and ramps it to -0.1 , because applying it from the start makes the policy learn only to hold the object still, and ViserDex [116], whose performance-based curriculum runs over regularisation weight, action latency and time between successes, fails completely without the regularisation component.

Population-based training appears once. DexPBT [111] trains populations of up to 32 agents, mutating the middle third of each population and replacing the bottom with mutated copies of the top, and reports 0.32 trillion environment steps for its largest run. The technique is routine outside robotics and nothing else in this corpus uses it.

A language model can write the reward outright, which removes the hand-tuning and replaces it with a search whose objective nobody has read. Eureka [16] has GPT-4 write the reward function directly, constrained to return a total and a dictionary of named components, and feeds per-component statistics back between iterations. Its appendix gives an example whose reward correlates at -0.26 with the human-written one and still scores 1.45 on the human-normalized metric, the strongest argument in the corpus that hand-tuned weights are not a ceiling. DrEureka [118] moves the safety requirement into the prompt instead of a penalty term, asking in words for a cube rotating at about 0.25 radians per second with fingers penalized for leaving their initial pose. Neither can be checked against a file. Eureka’s generated rewards are written at run time to a gitignored directory and never committed, so the reward behind any reported number cannot be recovered from the repository, though its README documents the paper’s budget of five iterations and sixteen samples as the default and marks the shipped one-iteration block a fast-test preset. DrEureka’s repository contains only the locomotion and globe-walking trees, so its LEAP-hand reward has no code at all.

Where reinforcement learning stalls. Reinforcement learning stalls on objectives that are not learnable as stated. AnyRotate [40] found the angular-velocity objective unlearnable in the multi-axis setting and replaced it with a moving keypoint target. RotateIt [72] reports that its multi-axis policy “does not converge when training with only reinforcement learning” and adds an imitation loss against single-axis oracles. Eureka [16] cannot spin a pen from scratch and needs a pretrain-then-finetune split, with the from-scratch ablation failing to complete one cycle. Section V-C’s contact-graph term exists because of the general shape of the problem: contact moves the object off the reference, so return goes down, so the policy learns not to touch the object.

It stalls on geometry too. Hora [76] fails on objects under 4 cm across because the fingers collide with each other. What it does not stall on is publication: the real-trial counts behind these headlines are small enough that Section VII-A treats them as the reporting problem they are.

C. Learning from human data

Teleoperation and retargeting. Thirty corpus rows describe a system for getting human motion onto a robot hand; the

repository lists them with the retargeting objective compressed to one clause each.

Retargeting splits four ways. The dominant family matches task-space vectors between the human and robot hands. DexPilot [119] set the pattern with a weighted squared distance between fingertip vectors, switching weights of 1, 200 and 400 by whether a pair is within threshold, a minimum-separation constant of 3 cm between primary fingers, and a regulariser toward the open hand, solved online with SLSQP; AnyTeleop [120], Open-TeleVision [121], Bunny-VisionPro [44] and ACE [122] all use the same formulation. The second copies joint angles where the kinematics allow it, as Holo-Dex [123] does for index, middle and ring while solving the thumb by inverse kinematics and discarding the pinky the Allegro does not have. The third learns the map offline: Geometric Retargeting [124] trains a per-finger network in three to five minutes against motion-direction, coverage, flatness, pinch and collision criteria, then runs at 1 kHz against the 60 to 100 Hz of online optimization, raising configuration-space coverage from 38 to 90 percent and one-time grasp success from 55 to 87.5 percent. The fourth removes retargeting from the loop: DexUMI [70] builds an exoskeleton whose fingertip workspace is optimized to match the target hand, then regresses encoder values straight to motor values.

What that tabulation does not carry is the point. Only 12 of the 30 rows record a latency at all, three of those give no number, and only 7 state a rig cost. Where costs are stated they are low and falling, from [123]’s \$399 headset through the \$600 exoskeleton of [122] and the \$600 haptic glove of [125] to the \$4,000 backpack of [126] and the \$6,000 rig of [45]. Latency is not one quantity: DexPilot [119]’s “about one second” and Holo-Dex [123]’s “under 100 milliseconds” are end to end, while Geometric Retargeting [124]’s 1 kHz and AnyTeleop [120]’s 9 to 10 ms retargeting are single stages and several cells give a control rate. That tabulation’s latency column mixes the two, so its spread is not a range. Only 13 rows report how many trajectories were collected, and only 7 report hours.

Where throughput is measured, human collection beats teleoperation by a wide margin. Qiu et al. [127] time a grasp demonstration at 3.79 seconds for a bare human and 19.72 seconds through a teleoperated humanoid, and a pour at 4.81 against 37.31 seconds. It attributes the gap to retargeting latency and to the workspace of a 7-DoF arm rather than to anything about the hand.

Imitation architectures. Four architectures cover the corpus. Action chunking came from [128], which predicts a chunk of joint targets with a CVAE and combines overlapping chunks by exponentially weighted ensembling, reaching 80 to 90 percent on fine bimanual tasks from about 50 demonstrations each. Diffusion came from Diffusion Policy [129], which denoises an action sequence and reports a 46.9 percent average success improvement over LSTM-GMM, IBC and BET across 15 tasks. 3D Diffusion Policy [130] swaps the image encoder for a small point-cloud encoder and reports 74.4 percent against 59.8 percent across 72 simulated tasks, and 85.0 against 35.0 percent on four real tasks from 40 demonstrations each. Flow matching is the 2024-onward default for large models and

carries 8 of the 112 rows. Discrete action tokens are the fourth, used by [25] and [131], and are the only one of the four that needs no continuous head at all.

Data generation is a separate lever and is underrated. DexMimicGen [46] turns 60 human source demonstrations into 21,000 simulated ones across 9 tasks and 3 embodiments, and a real Fourier GR1 with two Inspire hands reaches 90 percent from 40 generated demonstrations against 0 percent from the 4 source ones. Dex1B [132] iterates optimization, a CVAE proposal model and a simulator filter into roughly a billion synthetic grasps, and its CVAE baseline beats the prior best by 22 points.

Human video without a robot. Seven rows train a dexterous-hand policy from video with no teleoperation at any stage. DexMV [133] retargets 700 self-recorded demonstrations by matching palm-to-fingertip task-space vectors and estimating actions by inverse dynamics, VideoDex [134] mines 965 trajectories from EpicKitchens to pretrain a neural dynamic policy, and DexVIP [135] goes further into the reward, clustering 715 curated HowTo100M frames into one consensus grasp pose per object category and adding it as a reward term. OKAMI [136], Lum et al. [137] and Guzey et al. [138] work from a single video each, and Goswami et al. [139] pretrain a world model on 829 hours of EgoDex.

The embodiment gap. Three papers measure the gap directly rather than asserting it. OKAMI [136] runs the same reference-plan and warping pipeline with and without human body and hand retargeting, and the version without it loses 75.0, 42.1, 82.0 and 74.0 points across four task and setting pairs. That is the largest clean embodiment-gap number in the corpus.

Qiu et al. [127] ablate two mitigations separately on the same vertical-grasp task over 10 trials. The full method scores 4 out of 10. Removing the interpolation that slows human trajectories by a factor of four drops it to 1 out of 10. Removing the unified 54-dimensional state space while keeping the slowdown drops it to 0 out of 10, because the policy is handed a shortcut for telling the two embodiments apart. EgoZero [140] reports the sharpest version of the same effect. Removing 3D augmentation drops every one of its seven tasks to 0 out of 15, and substituting monocular depth for triangulated depth does the same, because the best metric depth models it tested carry more than 5 cm of error. It also quantifies what its hand-pose pipeline costs, at 1 to 2 cm of action-label error.

Chen et al. [141] supply the counterexample. Their method retargets only the wrist and lets reinforcement learning discover finger motion, and their ablation that adds a fingertip-matching reward “does not yield benefits and even leads to lower performance”. Richer human correspondence is not uniformly better, and this is the one result in the corpus that says so with an ablation.

Tracking a human reference with physics. Twelve method rows sit in the human-reference tracking family. Eight track a human hand on an object and are covered here; the other four sit at the edges, Lum et al. [137] tracking only the object’s trajectory, DexMan [142] retargeting bimanual video onto a full humanoid, and HumanPlus [27] and OmniH2O [28] tracking whole-body motion. A reference is retargeted onto the robot and a policy trained to make the simulated body

follow it: the reference supplies the shaping reward engineering would otherwise have to invent, and the simulator the physical consistency pure imitation lacks.

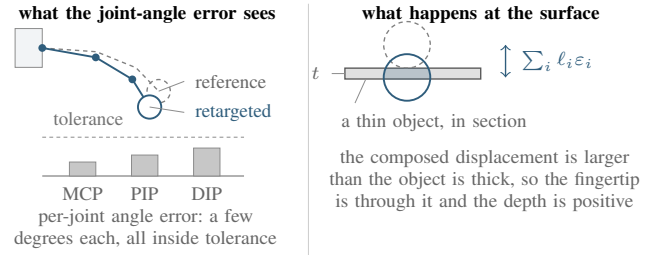
The multiplicative reward that most human-reference methods now use has a single source. PhysHOI [143] is the origin of the form. It multiplies a body term, an object term, an interaction-graph term and a contact-graph term, and reaches 95.4 percent success on GRAB against 27.0 percent for a DeepMimic baseline. The contact-graph term exists to stop the policy learning not to touch the object. Omnigrasp [12] replaces the action space with a pretrained latent motion prior and reports 94.6 percent grasp success and 84.8 percent trajectory success on GRAB, while stating outright that it omits penetration metrics.

The next step was to stop treating demonstrations as fixed and let the method generate its own. DexTrack [97] adds an imitation loss to PPO and mines its own demonstrations through a homotopy search over easier neighbouring trajectories, reaching 46.70 and 65.48 percent on GRAB at loose and strict thresholds against 38.58 and 54.82 for the best baseline. It has a penetration-depth formula and applies it only to input references, never to its own rollouts, and presents tolerance of “severe hand-object penetrations” as robustness. ManipTrans [18] freezes a generalist hand imitator and trains a per-task residual on top, reaching 58.1 percent single-hand and 39.5 percent bimanual success on OakInk-V2. Its stance on contact is to raise the friction coefficient above the real value rather than model skin deformation, and its real deployment is open-loop replay.

Two papers in this corpus ran the same curriculum and disagreed about whether it helps. DexMachina [19] moves to Genesis with 12,000 environments and six hand URDFs, and drives the object with virtual controllers that decay to zero. Its re-implementation of [18]’s curriculum does not beat no curriculum on its own setup, which is a direct disagreement between two papers worth quoting in both directions. Dexlore [75] drops the explicit retargeting stage and treats the reference as soft guidance, with termination thresholds derived from the failure rate, reaching 87.7 percent on GRAB with an Inspire hand.

TopoRetarget [144] is the only corpus method that quantifies penetration as an evaluation quantity, reporting maximum depth and the fraction of frames past 2 mm, and constraining it during retargeting with a 1 mm soft tolerance and a 30 mm hard bound. It reaches 87.5 percent on its own pen-spinning set against 46.9 for the best baseline. It does not re-measure penetration after the tracking policy runs, so the property it constrains is a property of the reference and not of the behavior. Fig. 8 draws why the usual retargeting objective cannot see the failure at all: joint-angle error is summed over joints and carries no term for the object, so errors small enough to pass at every joint compose along the finger into a fingertip displacement larger than a thin object is thick, and the pose is scored as good with the object inside the finger.

That is the pattern across all eight, and it is the reference-versus-rollout split in its sharpest form. Penetration is handled at the reference, if at all, and never at the rollout. Chen et al. [141] complete the set with the only real-robot numbers



The gap. Joint-angle error is a sum over joints and carries no term for the object, so it cannot see the right-hand panel at all. Errors small enough to pass at every joint compose along the finger into a fingertip displacement larger than a thin object is thick, and the pose is scored as good with the object inside the finger. Sec. V’s one method that measures penetration measures it on the reference, before the policy runs.

Fig. 8: A retargeted pose can be scored good and still be inside the object. The joint-angle error is a sum over joints with no term for the object; a few degrees at each joint compose along the finger into a fingertip displacement larger than a thin object is thick.

among them, from 100 percent on a microwave and a laptop down to 41.2 percent on a ketchup bottle over 20 trials each.

Cross-hand retargeting. Human data does not port across hands, and the map is usually left unstated. Fifty-three method rows use human data and name 44 distinct hand strings between them, and twenty of those appear among the 30 teleoperation systems with a stated retargeting objective. The other 33 never say how the human motion reached the hand, and the stated objectives do not converge, from a fingertip keypoint-vector energy in [120] to a per-joint regression from exoskeleton encoders in [70] to no retargeting at all in [140]. A shared benchmark could make these mappings comparable. TopoRetarget’s format [144] is a useful starting point: it reports contact precision, contact alignment, maximum penetration and the share of frames past 2 mm, per hand and per dataset.

D. Generalist and vision-language-action policies

The finding is the size of the hand. Eighteen rows carry the generalist tag. Fourteen of them settle whether the reported evaluation ran on a multi-fingered hand. Eight of those fourteen did, six report no hand result at all, and the remaining four never say, GR00T N1.6 [29] naming no end effector anywhere on its page.

Five of the eight state a hand size, and that comparison has to be made in actuated degrees of freedom rather than joints, for the reason Section III opens with. Four are six, the Inspire RH56DFX among them actuating six of its twelve joints, and one is twelve, [145]’s XHAND. Their median is 6. Two of the four rows that never settle the hand question do state a size, and both are large: 21 for [84]’s ByteDexter V2 and 22 for [146]’s Sharpa Wave. Of the 59 reward-learning rows, 48 state a count, 38 of those are 16 or above, and their median is 16: an Allegro and a LEAP actuate 16, and a Shadow actuates 20 of its 24 joints. So no row that settles the question reaches the band the reinforcement-learning literature of Section V-B works in, and the two rows that reach it on paper are the two that never say whether the hand was in the evaluation. An earlier version of this claim counted eleven rows as evaluating

on a hand, which mixed the settled eight with rows the notes leave open.

The six with no hand result are parallel-jaw throughout (π_0 [22], whose released code encodes each ALOHA gripper as one scalar, $\pi_{0.5}$ [23], $\pi_{0.6}^*$ [24], OpenVLA [25], Helix [83], which claims individual-finger control of a 35-DoF upper body with no success rate or trial count for any task, and Gemini Robotics [147], whose Apollo hand appears in a figure caption with no number attached). Its successor [26] closed that gap with Apollo progress scores, not success rates, of 0.74 down to 0.62 across five generalisation axes. DexVLA [49] is the borderline case the other way: a hand in one real-robot task family, grippers in the other five and in all 91 pretraining tasks.

The released code also drifts. GR00T N1 [148]’s repository is a later N1.7 generation with a different backbone, a different action horizon and embodiments the paper does not describe, and the $\pi_{0.5}$ repository [23] supports only the flow-matching head, not the hybrid recipe the paper describes.

Dexterous hands in generalist policies. The gap is size rather than absence. The generalist policies that do touch a hand run it at roughly a third of the actuation the reinforcement-learning literature assumes, and the rows that never settle the question are the smaller half of the same gap. The standard generalist suite, with the hand’s actuated degrees of freedom and its vendor printed beside the number, should include at least one dexterous-hand task.

E. Model-based baselines and hybrids

Three corpus methods solve dexterous tasks without learning a policy, and they are the control group the rest of this section lacks.

Not every method here learns a policy. A smaller line plans through a model instead, and reaches dexterous tasks on a fraction of the data. PDDM [109] learns an ensemble of dynamics models and plans through it with an MPPI-style optimiser, replanning every step. It needs 1 to 2 hours of data in simulation and 2 to 4 hours on a real 24-DoF Shadow Hand. Predictive Sampling [149] removes the learned model too and samples ten rollouts per step through MuJoCo itself, planning in 1 to 20 milliseconds. It reorients a cube with a Shadow Hand in real time from scratch, and reports no success criterion, no trial count and no real-robot result.

The work of Pang et al. [36] is the most substantial of the three. It proves that the randomised smoothing implicit in reinforcement learning and an analytic log-barrier relaxation compute the same local linear model of contact, then uses the analytic version inside a trajectory optimiser and an RRT. Allegro in-hand rotation takes 19.59 seconds to optimize and its plate-pickup task 117.16 seconds of planning on a 16-core CPU, against the GPU-days of Section V-B, and it is the only method here that imposes non-penetration as a hard constraint. Its own limitation is the honest part: its 3D systems transfer to hardware far worse than its 2D ones, because the quasi-dynamic assumption breaks and planned grasps miss contacts under a second-order solver.

Hybrids, and where the field has converged. The recurring shape is reinforcement learning in simulation distilled into

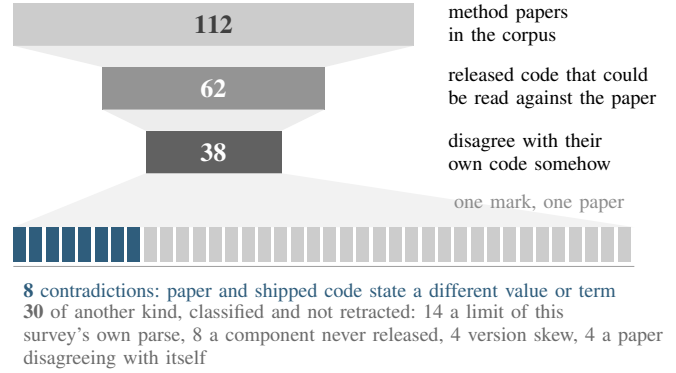


Fig. 9: Papers and their own released repositories state different things, and the funnel ends at the 8 the survey stands behind as contradictions. Table II names them, with the file and both values.

a policy that looks like an imitation policy, and it appears in four variants. The first distills a privileged teacher into a vision student inside one paper, which is [76], [17], [72], [71] and [116]. The second uses reinforcement learning as a demonstration factory: DexTrack [97] mines demonstrations with per-trajectory RL and trains a generalist on them, ManipTrans [18] does the same to build a 3.3K-episode dataset, and DexterityGen [150] trains a diffusion controller on ten billion simulated grasp-to-grasp transitions and projects a human teleoperator’s coarse command onto it, taking four reorientation tasks from 0 out of 20 under raw teleoperation to 12, 13, 10 and 9 out of 20. The third generates data without reinforcement learning at all, which is [46], [132] and [151]; Dex1B’s row is classed as a dataset, so Fig. 7’s leaf, which counts method rows, holds the other two. The fourth wraps an RL policy in a residual, which is [152] at 88.8 percent over 3,200 objects in 12 GPU-hours.

In the variants that dominate 2025 and 2026 work, the reward is no longer the objective of the deployed policy. It is the objective of the process that produced the deployed policy’s training data. Reward engineering has moved upstream into data curation, and every reward-shaping pathology in Section V-B now reaches the shipped policy through a dataset rather than a gradient.

F. Paper against released code

Thirty-eight of the 112 method rows record a disagreement between a paper and the code it released, and all 38 released code, so they sit inside the 62 rows that released anything (Fig. 9). They are not one kind of thing. Eight are contradictions, where paper and code state different values or different terms. Fourteen are limits of this survey’s own parse, where the body or config that would settle the question was never recovered and the row says so. Eight released code without the described component in it, four are version skew against a later repository, and four are a paper disagreeing with itself. The fourth version skew is [29], which ships a main branch one generation later than the checkpoint its page describes.

Eight is the number to quote, seven at high confidence and one, PenSpin [73], held at medium. Eight of 62 is 13 percent, and it is a floor rather than a rate: the census covers method

TABLE II: The 9 rows where a paper and its released repository state different values, at the commit fetched, with confidence.

method	conf.	repository, fetched commit	file in it, and where	what the paper prints	what that file contains
PDDM [109]	high	google-research/pddm 06b88cdebaf	pddm/envs/cube/cube_ env.py _get_obs	observation dimension 46 for in-hand reorientation, Table 2	the six fields <code>_get_obs</code> concatenates sum to 39 on the code's own inline comments
DexPoint [38]	high	yzqin/ dexpoint-release 17f1e238bb	dexpoint/env/rl_env/ relocate_env.py AllegroRelocateREnv. get_reward	a four-term reward, Eq. 5	<code>get_reward</code> sums nine, among them 1.0 / (0.06 + <code>finger_ object_dist</code>) and <code>controller_penalty</code>
DeXtreme [15]	high	isaac-sim/ IsaacGymEnvs aeed298638	isaacgymenvs/cfg/task/ AllegroHandDextremeADR. yaml actionDeltaPenalty Scale	action-delta penalty weight -0.25, Table 2	<code>actionDeltaPenaltyScale</code> : -0.2, and -0.01 in the ManualDR yaml beside it
Visual Dexterity [17]	high	Improbable-AI/dexenv ad9634e9d2	dexenv/conf/dclaw.yaml alg.num_envs	32,000 teacher training environments, Table S1	<code>num_envs</code> : 8000, and 16384 in the parent <code>hand_default.yaml</code>
DexPBT [111]	high	NVIDIA-Omniverse/ IsaacGymEnvs aeed298638	isaacgymenvs/tasks/ allegro_kuka/allegro_ kuka_base.py compute_kuka_reward	four staged reward terms, Sec. III-C	eight summed components, <code>hand_ delta_penalty</code> multiplied by 0 with the comment currently disabled
PhysHOI [143]	high	wyhuai/PhysHOI 6095c605e2	physhoi/env/tasks/ physhoi.py compute_humanoid_ reward	object-rotation weights 0.1 and 0.01 for GRAB, Table 4	<code>eor</code> and <code>eorv</code> set to <code>torch. zeros_like(ep)</code> , the computation commented out on the same lines
UniDexGrasp [153]	high	PKU-EPIC/UniDexGrasp 36c9bfcf7c	dexgrasp_policy/ dexgrasp/tasks/shadow_ hand_grasp.py compute_hand_reward, goal_cond branch	seven named weights from 0.1 to 10, Table 7	a <code>torch.where</code> cascade on literals -0.5, -1.0, 0.9, 0.1, 0.2, none of the seven present
PenSpin [73]	medium	HaozhiQi/penspin 5035c52dc9	configs/task/Allegro HandHora.yaml forceScale	a disturbance force of 0.2 times object mass at probability 0.25, Table 8	<code>forceScale</code> : 0.0 and random <code>ForceProbScalar</code> : 0.0, the only shipped config naming either five summed terms; <code>_compute_ energy_reward</code> ends return 0 and <code>_compute_fingering_ reward</code> return 0.0
PianoMime [32]	high	sNiper-Qian/ pianomime c4abefac8d	single_task/piano_ with_shadow_hands_res. py _set_rewards	two weighted terms, 2/3 and 1/3, Table 3	

rows only, so RoboPianist [30], whose row is a benchmark, does not enter it although it sums five reward terms against the three its Table 2 documents. It is also a screened count rather than a first pass. Sixteen rows were drawn at first and eight have since left the class: six under adversarial re-reading, one, OmniH2O [28], once writing to its authors sent someone back to the evidence, and one, DexPoint [38], because its corresponding author replied and showed the survey had misread the code, the term it called a departure from the paper’s formula being the paper’s formula, with the line defining it absent from this survey’s own parse. Two more were narrowed and stayed, and each revision is recorded in the row it concerns beside the comparison it revises. That screening is evidence that the eight were checked, not a bound on how many a fuller reading would find. Among the 21 reorientation methods scored term by term, seven released a repository and four of those state something different from their paper, the other three being a missing environment, a later generation and a default-value question their own READMEs settle.

PhysHOI [143] is the clearest example. Its paper assigns nonzero weights to object rotation and rotation velocity on the GRAB dataset. In the public implementation, both errors are set to zero and the calculations are commented out. The paper does specify zero weights for the separate BallPlay task, but the implementation does not condition this choice on the dataset. Its reported success criterion uses position alone, so that score cannot expose the difference. The claim here is limited to this conflict between the paper and the public revision listed in Table II; it does not establish which code produced the published experiments.

The other cases vary in importance. DeXtreme [15] gives one action-change penalty in the paper and a different value in each of two public configurations. PianoMime [32] describes two reward terms, while its implementation sums five and forces two of them to return zero. Table II gives the exact file, revision, and values for these and the remaining six cases, where they can be checked without interrupting the argument with source-code identifiers.

Scope and correction. The authors of the nine works were written to on 19 September 2026, before this survey was posted. Each letter quotes the claim, the repository, commit and file behind it, and the sentences this survey would print, and asks three things: whether the reading is right, whether there is a reason the released configuration differs from the paper’s, and whether the authors want the wording changed. What a reply can add is what the artifacts cannot give: which shipped configuration belongs to which stage of a pipeline, whether the fetched commit is the one behind the reported numbers, and whether an untagged branch holds the code that was run. Each was given until 10 October 2026, and told that silence would be recorded as silence rather than as agreement. This version was written between those two dates: no reply had been received at the time of writing, no row here has been changed by one, and a window that closes in silence will be recorded in the next version as silence and not as assent. The comparison is nonetheless limited to two public artifacts: the paper and the repository revision named in Table II. It makes no claim about intent or about the private code used for the reported experiments: a fetched commit is not the code that produced a paper’s numbers. It may predate, postdate, or differ from

that code. Hora [76] says exactly that about itself by directing readers to a tagged release rather than its default branch. The letters are in `outreach/`, with the address each went to and its provenance in `outreach/RECIPIENTS.md`, and any reply, with its effect on the row it concerns, is recorded in that row’s `mismatch_review`. Every comparison is correctable in public through the address in the author block; supporting evidence changes the record, the derived counts, and the next version of the text.

The classification distinguishes a term that is absent, present with zero weight, or assigned a different weight. In 14 cases the parsed repository cannot settle the comparison, and those cases are recorded as limitations rather than conflicts. Appendix C prints all 38 row by row in their five classes, each with the file, the value on both sides, the recorded confidence, and the review note where a comparison was narrowed or withdrawn.

Review and correction. This is a result about publishing practice in robot learning rather than about dexterous manipulation in particular: the dexterous corpus is its sample, not its subject. Eight is a lower bound because 46 method papers released no code to inspect and four comparisons remain unsettled. It is also the result of a conservative review. The first draft identified sixteen conflicts; eight have since left the count and two more were narrowed and remain in it. Six failed under adversarial rereading, and a seventh was reclassified when its apparent weight mismatch proved consistent with a curriculum factor. PenSpin [73], the sole medium-confidence case retained, concerns a disabled disturbance rather than the tactile input claimed in an earlier draft. Each change and its reason is printed beside the comparison it revises in Appendix C. Future papers can avoid most of this ambiguity by generating the reward table directly from the configuration used for training and citing the corresponding revision.

VI. BIMANUAL DEXTEROUS MANIPULATION

A warning first. Fifty-three corpus method rows carry the two-hand flag, and that flag says only that the robot has two end effectors. Thirteen of the 53 put no dexterous hand on the robot at all. ALOHA, RDT-1B, EgoMimic and H-RDT [128], [154], [155], [156] and UMI [157] are parallel-jaw throughout; π_0 , $\pi_{0.5}$ and $\pi_{0.6}^*$ [22], [23], [24] and Diffusion Policy [129] name no hand and run grippers on every embodiment; Gemini Robotics [147] and Gemini Robotics 1.5 [26] report every bimanual number on grippers and show five-fingered hands only qualitatively; and Helix [83] and GR00T N1.6 [29] name no end effector anywhere on the page. Three more run a gripper on one embodiment and a hand on another: ACE and 3D Diffusion Policy [122], [130] and Open-TeleVision [121], whose Unitree H1 carries 6-DoF Inspire hands and whose Fourier GR-1 carries a 1-DoF jaw.

The denominator for this section is 28: the method rows whose notes place a learned closed-loop controller on two multi-fingered hands. It is the 53 less those 16; less two static grasp-synthesis methods, BimanGrasp [42] and BiDexGrasp [158]; less three systems with no learned policy, Castro et al. and DexTeleop-0 [93], [69] and Pang et al. [36]; and less four rows where no learned policy holds both hands:

OmniH2O [28], whose fingers are mapped open-loop from the Vision Pro and sit outside its 19-DoF policy; OKAMI [136], whose headline pipeline is open-loop retargeting with a learned policy only in a side experiment; DexDeform [33], a skill model refined by trajectory optimization rather than a closed-loop controller; and Omnigrasp [12], a simulated human body with no bimanual task. Frau-Alfaro et al. [21] never enter, because their row records one hand. Benchmarks and datasets are outside the 28 by class, Bi-DexHands and Bench2Dex [47], [88] and RoboPianist [30] being benchmarks and RP1M [31] and HumanoidGen [159] datasets; they are quoted here as evidence and never counted. Every paper below runs two multi-fingered hands unless said otherwise.

A. Coupling, roles and collision

When both hands hold the same object, the object closes a kinematic loop between them. Neither hand can move without changing what the other must do. BiDexGrasp [158] reports that the coupled objective “often yields imbalanced solutions, where one hand dominates stability while the other contributes marginally.” ArtiGrasp [160] reports simulation speed scaling “roughly quadratically with the number of contacts,” and trains each hand alone before pairing them.

Role asymmetry is an assumption most of these systems make silently. BiDexHD [161] states it outright: “we assume the robot to be right-handed by default, i.e., the left hand handles the target object and the right hand handles the tool,” and Lin et al. [162] bake the same split into their reward, putting reference contact keypoints on the bottle base for the left fingertips and on the lid for the right. The bias is in the data first: TACO [163] recruited 14 right-handed subjects and measures the right hand moving consistently faster.

The arms collide, and the corpus handles this by construction rather than by control. RoboPianist [30] ships a forearm-forearm collision term that its own reward table never lists. Two papers name arm collision as unsolved. DexMimicGen [46] “does not explicitly handle collisions,” and DexMan [142] says its control parameterisation “overlooks full arm posture,” which it needs to avoid them.

The action space doubles and the observation grows faster. Bi-DexHands [47] uses a 52-dimensional action for 18 of its 20 tasks against observations of 398 to 446 dimensions. GR-Dexter [84] emits an 88-dimensional chunk of arm joints, end-effector poses, hand joints and fingertips. What the larger vector does not carry is any statement of who does what. The usual answer is one scalar reward summed over both sides, and the cost is a compounding failure rate. ManipTrans [18] scores success only if both hands succeed, and its OakInk-V2 success rate falls from 58.1 percent on single-hand sequences to 39.5 percent on bimanual ones.

B. Coordination architectures

Fig. 10 sets the four mechanisms side by side. Every row of the 28 carries exactly one architecture, read from its note: one policy over both hands, a network per hand, leader and follower, or a note that does not say. The fourth panel is not a

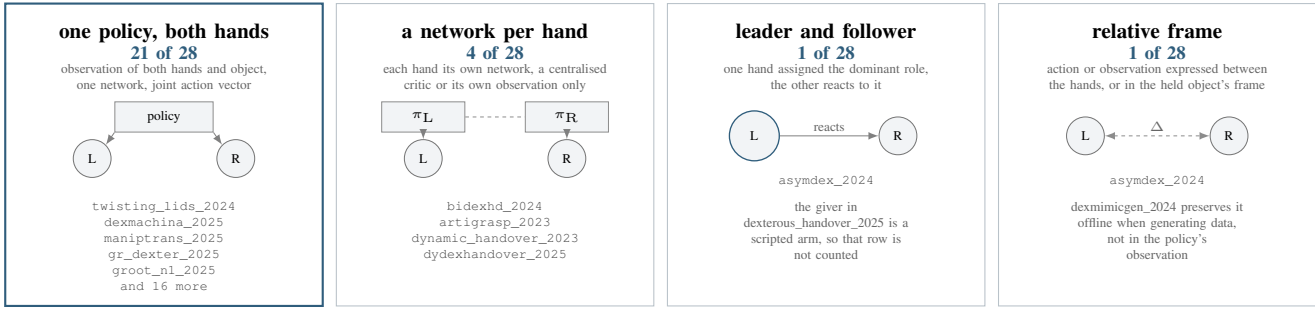


Fig. 10: Four mechanisms for controlling two dexterous hands, counted over the 28 corpus rows with a learned closed-loop controller on two multi-fingered hands, each read from its note. The panels are not a partition: the fourth is the same single row as the third, and the two rows whose notes never say which architecture they use appear in neither.

fourth architecture but the relative frame, and it is the same single row as the third.

Twenty-one of the 28 put one policy over both hands, which is three quarters of the set: the trackers and tracking-adjacent methods [19], [142], [18], [141] and [27]; the reinforcement-learning tasks [111], [162], [16], [32] and [108]; the demonstration pipelines [45], [126], [164], [46], [51] and [127]; and the generalist policies [84], [148], [145], [131] and [146]. In every one of them a single network takes a concatenated two-hand observation and emits a two-hand action. Two rows do not say which they are, Bunny-VisionPro [44] and DexImit [151], whose notes describe the rig and the data pipeline but never the policy’s own decomposition. The concentration is not the outcome of a comparison that was won.

Four of the 28 give each hand its own network, and the two papers that compare the choice disagree. Bi-DexHands [47], a benchmark row and so outside the 28, ships the MARL baselines and finds PPO over the full observation beats HAPPO and MAPPO “in most cases,” with the gap narrowing on tasks that need both hands, because PPO “can use all observations” where MARL sees only part. BiDexHD [161] concludes the opposite. Its independent PPO teachers score 74.59 percent stage-two tracking rate on trained tasks against 53.88 for a centralised policy over both observations. Both are simulation only, on different tasks and hands, so neither settles it. ArtiGrasp [160] trains one PPO policy per hand, and Dynamic Handover [20] and DyDexHandover [43] use MAPPO with one agent per arm-hand system.

One of the 28 assigns explicit leader and follower roles, and the same one is the only policy expressed in a relative frame. AsymDex [41] is that paper, and the only corpus paper that puts either mechanism inside a policy. It gives the dominant hand full finger and wrist control, restricts the facilitating hand to a 6-DoF base pose, and writes the dominant hand and the object in a frame attached to the object the facilitating hand holds. That cuts the observation from 176 dimensions to 88 and the action from 52 to 26. It ablates the two mechanisms separately, which no other corpus paper does. On Block in cup the full method scores 0.7701 over five seeds, against 0.1086 for relative frames without asymmetry and 0.0164 for asymmetry without them. Twist Lid transfers zero-shot at 18 of 20 real trials. The real rig pairs a 16-DoF Allegro with a 6-DoF Ability Hand because only one Allegro was available,

so the roles are confounded with the hardware. Two other rows use a relative quantity without giving either hand a role: DexWild [164] appends the inter-hand pose to its observation, and DexMimicGen [46] applies one shared SE(3) transform to both arms’ source segments when generating data, which is the mechanism offline rather than in a policy.

Evidence still needed. One architecture carries three quarters of the field, and the choice behind it has been compared twice, with opposite outcomes, and ablated once. A decisive comparison would run one handover task on three architectures with the same hand and the same seeds, reporting separate giver and receiver returns rather than a single shared one.

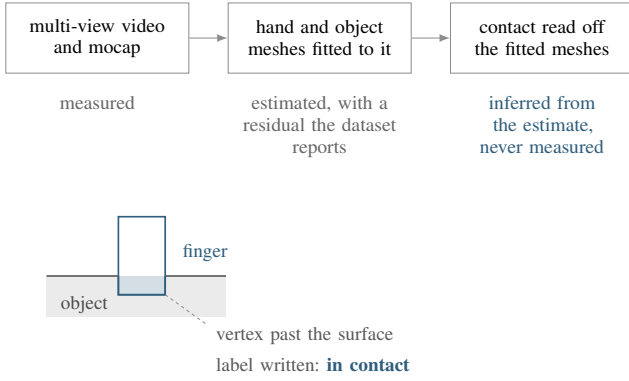
C. Benchmarks, datasets and the shared axes

Two purpose-built bimanual dexterous suites exist in the corpus, four years apart.

The two bimanual benchmarks in the corpus trade coverage against measurement, in opposite directions. Bi-DexHands [47] is 20 tasks on two Shadow Hands in Isaac Gym, ordered by the infant age at which humans acquire the skill, at 2048 environments and a reported 30,000-plus FPS. Its measurement discipline is weaker than its coverage. It reports reward and normalized score, never a success rate. Its only success flag in code tests the object-to-goal distance against 3 cm, which ignores orientation and exists only in the four catching tasks. Any success rate later work attributes to Bi-DexHands comes from that flag or its own definition.

Bench2Dex [88] is the more instrumented of the two. It runs 26 long-horizon tasks in Isaac Lab across 12 arm-and-hand embodiments, with roughly 1.3K teleoperated demonstrations in eight modalities, including a ray-cast tactile image that maps all 12 hand geometries into one representation. Success needs the terminal predicate held for 0.5 s. Matched, GR00T N1.5 leads at 631 of 1300 rollouts, 48.5 percent, against ACT at 29.5, $\pi_{0.5}$ at 27.3 and Diffusion Policy at 12.9. Under combined perturbation those first two compress to 19.8 and 19.7. Tasks are not crossed with embodiments, so no contrast isolates the hand, and the authors state the hand and tactile models are not calibrated against matching physical hardware.

Ten corpus rows are bimanual datasets, and five are two-hand interaction capture. ARCTIC [165] gives 339 mocap sequences of 11 one-DoF articulated objects held in both hands, and deliberately labels over-shooting vertices as in contact, so its



The annotation is as good as the fit and no better, and a penetration is recorded as a contact rather than as an error. Of the five bimanual capture sets in Sec. VI, one publishes a penetration number for its own annotations.

Fig. 11: Where a capture set’s contact annotation comes from. Nothing in the chain is measured at a surface, and a vertex that has gone through the surface is written down as a contact rather than as an error.

ground truth records interpenetration as contact; Fig. 11 is the chain that produces it. TACO [163] gives 2.5K bimanual tool-use sequences fitted markerlessly under an attraction loss and a penetration loss the paper says trade against each other. OakInk-V2 [166] is the only corpus dataset publishing a penetration number for its own annotations, a mean depth of 0.25 cm over its 627 long-horizon sequences. HOT3D [167] gives 833 egocentric minutes with mocap-grade poses for both hands and up to six objects, and GigaHands [168] gives 2,034 markerless minutes from 56 subjects, neither with a penetration metric. These five feed the tracking work here: DexMachina [19] and ArtiGrasp [160] from ARCTIC, BiDexHD [161] from TACO, ManipTrans [18] from OakInk-V2 and DexMan [142] from both.

Methods on the same axes. The method tabulation carries the bimanual papers on the same columns as every other method here, and three of those columns are worth reading together.

Real trials. AsymDex [41] reports 20 per task, HATO [51] 10 per condition, DexMimicGen [46] 20 on a Fourier GR1 and BiDexGrasp [158] 260 across 30 objects. ManipTrans [18] reaches real Inspire hands by open-loop replay with no trial count and no success rate. Seven of the 28 have no real robot at all: [160], [161], [19], [142], [111], [43] and [32]. The two piano rows outside the denominator, RoboPianist [30] and RP1M [31], are simulation-only as well.

Penetration, and Fig. 12 is why it has to be reported beside a success rate rather than instead of one. Most bimanual rows read “not addressed.” BimanGrasp [42] gates on it, failing any grasp whose total penetration across hand-object, self and inter-hand exceeds 1.5 mm, and reports that “penetration remains the primary cause of grasp failure.” BiDexGrasp [158] reports it as a number, 0.15 to 0.20 cm maximum depth for its own grasps. DexImit [151] carries a named hand-hand penetration term in its grasp-synthesis objective. DexMachina [19] resolves it once, replaying retargeted joints against a fixed object “to eliminate object penetrations” and measuring nothing during

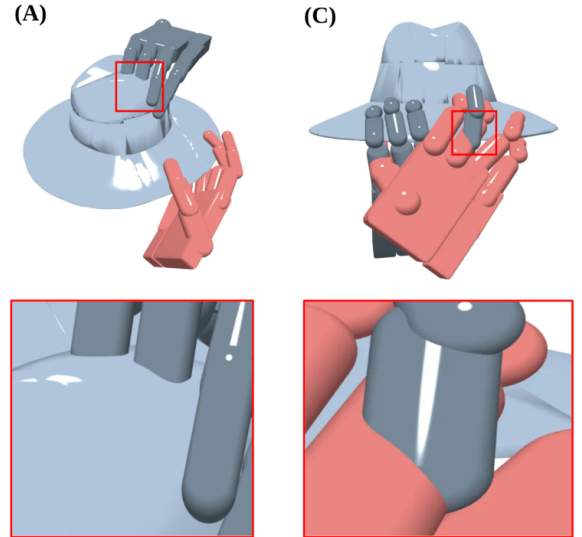


Fig. 12: Two failures of synthesised two-hand grasps, each enlarged at the contact: (A) a finger inside the object, (C) a finger inside the other hand. Both look plausible whole [42].

Fig. 9 of Shao and Xiao, arXiv:2411.15903, CC BY 4.0; panels (A) and (C) of Fig. 9.

rollout. ArtiGrasp [160] declines the metric outright, because its baselines “include a physics simulation which exhibits no interpenetration.”

Comparability. BimanGrasp [42] reports 54.03 percent success in Isaac Gym at friction 3. BiDexGrasp [158] re-runs the same grasps in MuJoCo at friction 0.6 and gets 26.80 percent, at 1.52 cm penetration depth. Neither number transfers, and Section V-C’s re-implementation result is the same lesson on the training side.

D. Handover, and what transfers from one hand

Handover is the one bimanual task where the hands are unambiguously asymmetric, because one gives and one receives. All three corpus handover papers use a single shared reward across giver and receiver. None defines separate objectives for the two roles.

Whether the two hands optimize the same objective is usually left implicit. Dynamic Handover [20] states one formula, $r = r_{\text{dis}} + r_{\text{linvel}} + r_{\text{torque}}$, and never says whether thrower and catcher receive different decompositions of it. DyDexHandover [43] is explicit that they do not: “Both hands share aligned objectives, forming a fully cooperative relationship,” with one weighted sum feeding both agents. Only the thrower gets an extra KL regulariser toward a policy pretrained on human throws, which constrains style and not role.

The third case is weaker still. In [21] only the receiver is learned. The giver is a UR5e with an Allegro hand that “holds the object without moving during the whole episode.” There is one agent, one reward and no second policy, its corpus row accordingly records one hand, and it is one of the rows the 28 excludes. The 94 percent often attached to this paper needs its conditions. It is Total Success, which counts “Indetermination” cases where the simulator failed to resolve collisions and the

object clipped through the giver’s hand, on the short prism, in simulation, over 100 episodes, with no real robot in the paper.

Nothing in the handover literature measures contact quality. Contact appears only as a positive signal, a boolean per-phalange touch in [21] and a boolean contact reward in [43]. The two systems that do handover well sidestep the problem. HATO [51] reaches 10 of 10 on a real slippery handover by imitating teleoperation with fingertip touch sensing and no physics model. Lin et al. [108] stage their reward with a discrete variable switching which hand’s fingertips are scored, and still calls handover its hardest task at 52.5 percent real success.

What transfers from single-hand work. The training recipe transfers intact. PPO with thousands of parallel environments, an asymmetric critic on privileged state, domain randomisation and distillation into a vision policy is the same in [162] and [108] as in Section V-B. Reward forms transfer literally, and the contact-goal reward $r_{\text{contact}} = \sum 1/(1 + \alpha d)$ appears in both.

Multi-task learning does not transfer, and the bimanual benchmarks show it clearest. Bi-DexHands [47] finds multi-task PPO and ProMP fail on MT4, MT20, ML4 and ML20. RoboPianist [30] finds F1 on the shared training song drops “from roughly 0.7 F1 for 1 song to almost 0 F1 for 16 songs,” with no positive transfer “regardless of the size of the pre-training tasks.”

Single-hand evaluation does not transfer either. Success criteria compound, because ManipTrans [18] fails a trajectory “if either hand fails to meet these conditions.” And object-centric metrics cannot say which hand caused the failure. BiDexHD [161] scores the steps where both objects track their references, and DexMachina [19] averages ADD across object parts before an AUC. Both say whether the task worked and nothing about which hand lost it.

One failure mode has no single-hand counterpart. Two hands can penetrate each other, and the corpus almost never looks. Three papers carry an inter-hand penetration term, and all three are static grasp synthesis: [42], [158] and [151]. Not one learned bimanual controller in the corpus measures or penalizes hand-hand penetration during a rollout. PianoMime [32] ships a flag that turns inter-hand collision off in the physics.

VII. EVALUATION, AND A PROTOCOL FOR COMPARISON

A. Reporting practice

Of the 112 method rows in the corpus, 89 report a real-robot experiment, 22 do not and one row is unsettled, which is 80 percent of the 111 the note settled. Among those 89, 70 state how many real trials produced the headline number, 79 percent of them. The 89 is the denominator that belongs to this statistic: the 22 rows with no real robot cannot state a real trial count, and counting them as silent turns a definitional impossibility into a reporting failure. Thirty-nine rows state a count of unseen test objects, 35 percent. Ninety-eight state how a rollout is scored, 88 percent. Sixty-two released code and 46 did not, with four rows unsettled, 57 percent of the 108 the note settled. Fig. 13 draws these six shares, each against the denominator that belongs to it. Every bar is a lower bound,

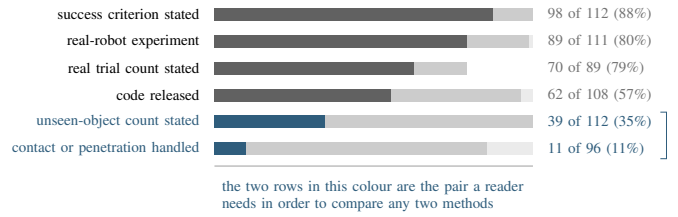


Fig. 13: What gets reported, and what does not. Each bar is a count of the 112 corpus method rows; the pale track behind it is the denominator that statistic is quoted against, printed beside the count, because four of the six are not 112. A faint tail past a track is the rows whose notes left that question unsettled. The two rarest reporting practices are marked in color.

for the reason the next paragraph sets out. Two of the six are marked, because they are the pair a reader needs in order to compare any two methods: without a count of unseen objects a success rate has no generalisation denominator, and without contact or penetration handling it says nothing about whether the hand passed through the object. They are also the two the field reports least.

These are counts of what this survey captured, not of what papers reported, and every one is a floor. A structured row holds a scalar. A paper that reports ten trials on each of nine tasks, or a scoring rubric instead of a threshold, or a count spread over four tables, has nothing the extraction can reduce to one integer, so it produces a null, and a null is then indistinguishable from a paper that said nothing. The bias runs one way: every miss converts a reporting paper into a silent one, and the survey’s argument is that the field reports badly, so the artifact flatters the argument. Section V-F makes the same disclosure about the paper/code count, subtracting the 13 disagreements that are limitations of this survey’s own parsing before declaring which number to quote, and the coverage statistics above need it more.

The nulls behind four of those six shares were therefore audited by hand: the trial counts, the success criteria and the unseen-object counts against the notes they came from, and the penetration field against the parsed sources themselves. Those four numbers above are post-audit. Appendix A gives what the audits recovered and what they could not: the counts moved by tens of rows, 19 trial counts and 4 criteria remain genuinely unsettled, and every bar in Fig. 13 is still a lower bound. The other two shares are not post-audit. The code-release field has never been audited this way, so its share is a floor of unmeasured depth, and the real-robot field’s one unsettled row is left out of its denominator rather than counted as silence.

The remaining gap is the one that matters. Nineteen of the 89 papers with a real robot never say how many times they ran it. A percentage with no denominator cannot be given an interval, so it cannot be compared with anything.

Where the denominator is stated it is small, and it is not one quantity. Some stored counts are per-cell, meaning ten trials on each task, or twenty per condition, or five per object. Others are grand totals over every cell. The rows now record which of the two each value is, and the two distributions are quoted separately. The 39 per-cell counts run from 5 to 100 with a median of 15 and quartiles at 10 and 20. The modal cell is

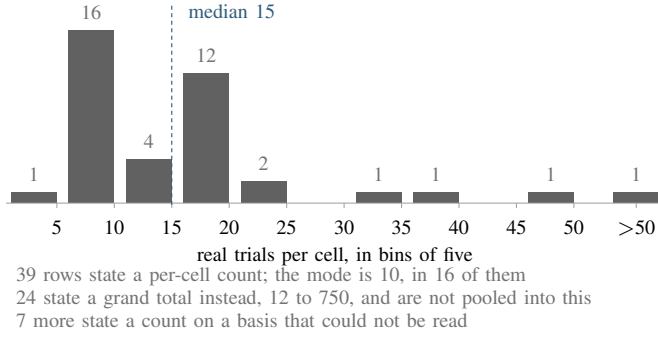


Fig. 14: Where a denominator is stated it is small: the 39 per-cell real-trial counts, binned by fives, median 15 marked. The 24 grand totals are a separate quantity, counted beneath the axis rather than pooled in.

10 trials, in 16 rows, then 20, in 12. The 24 grand totals run from 12 to 750 with a median of 110. Pooling the two gives a median of 20 and a range of 5 to 1287, and that pooled figure is the one an earlier draft of this section quoted. It describes nothing, because Hora [76]’s 240 and Visual Dexterity [17]’s 20 are experiments of comparable size recorded on different bases. Seven further counts could not be assigned a basis at all. Fig. 14 is the per-cell distribution.

The per-cell count is the one a Wilson interval attaches to. For s successes in n trials, with $\hat{p} = s/n$ and $z = z_{1-\alpha/2} = 1.96$ at the 95 percent level, the bounds of that interval are

$$\frac{2n\hat{p} + z^2 \pm z\sqrt{4n\hat{p}(1-\hat{p}) + z^2}}{2(n + z^2)}, \quad (1)$$

which, unlike the normal approximation, stays inside $[0, 1]$ and does not collapse to zero width at $\hat{p} = 0$ or $\hat{p} = 1$. At 20 trials a reported 60 percent carries a 95 percent Wilson interval of 39 to 78 percent, and a reported 80 percent an interval of 58 to 92 percent. Overlapping intervals are not a test, so the point is made with one: 12 of 20 against 16 of 20 is $z = 1.38$, $p = 0.17$, and the difference is not established. Two methods separated by 20 points at 20 trials, the larger of the two modal cell sizes, are not separated at all.

Reporting has improved a little and the experiments have not. About two thirds of papers from 2025 onward state a trial count, against three fifths before, but the counts themselves have not moved: the median has sat near twenty trials per cell every year since 2024, and earlier years rest on too few observations to read as a trend. A field can get better at disclosing a number without the number itself getting any better.

The denominators also sit on different hardware. The 103 method rows that name their own hand give 78 distinct hand strings between them, and this survey applies no normalization to those strings, so 78 is a count of strings and not of hand designs. Matching on the string, Allegro appears in 35, Shadow in 21, Inspire in 19 and LEAP in 12. A success rate on a 16-degree-of-freedom Allegro and a success rate on a 6-actuator Inspire hand are not measurements of the same thing.

Nor are the criteria. DexVerse [89] counts PickCube a success when the cube is “lifted at least 0.20 m above its resetting height”. Bench2Dex [88] requires its terminal predicate to hold for a continuous dwell time of 0.5 s, to reject transient

contacts. The Colosseum [169] counts an episode successful “if the model completes the task fully”. DexTrack [97] reports every success rate as a pair under two threshold sets, which on GRAB gives 46.70 and 65.48 percent for the same rollouts. Of the 14 rows that still state no criterion, ten have no success predicate at all. They report radians rotated or time-to-fall and never define a success, which is a fact about the paper rather than a gap in this survey. And four are unsettled by the note. Of the 19 criteria the audit recovered, all but one are a rubric, a staged partial credit, a judgment by eye or a deferral to a benchmark’s own definition rather than a threshold, and exactly one, [24], states a verbatim numeric one; Appendix A counts the four kinds. A rubric is a milder failure than silence and a worse one than a threshold, because it is reproducible inside a lab and not across two.

The axes that matter. Seven quantities dissociate in the published data, so they have to be reported separately.

Task success. Binary success discards the difference between near-misses and inaction. Snyder et al. [170] put it plainly: “a policy that completes 90% of the task is clearly better than a policy that is frozen the whole time, yet their success rates would be identically 0%”. In [171] policy C scores 17 percent overall on the pancake task while picking up the spatula and flipping the pancake in 23 of 23 attempts.

Robustness to perturbation. The Colosseum [169] measures a 30 to 50 percent success drop under single perturbation factors and at least 75 percent under all 14 together. In [88] GR00T N1.5 leads the matched condition at 48.5 percent and falls to 19.8 percent under combined shift, while $\pi_{0.5}$ goes from 27.3 to 19.7 and retains the most at 72.1 percent. The ranking at the anchor is not the ranking under shift.

Generalisation to unseen objects. Only 39 rows state a count and the median is 11 objects.

Physical plausibility of the contact. Eleven of the 96 rows whose contact handling the note settled address it, 11 percent, with 16 rows unknown. Section VII-B takes them apart.

Sample and wall-clock cost. Thirty-one of 112 rows state a parallel environment count and 18 a simulated episode count. RoboPianist [30] is the exception, at 5 million samples per song and roughly 5 hours per run on four Tesla K80 GPUs.

Real-robot transfer. Simulated rank order is not real rank order. AutoEval [172] scores Open- π_0 on put-eggplant-in-sink at 6 of 50 in SIMPLER and 47 of 50 on the real WidowX. Badithela et al. [173] state the limit directly: “the simulation-to-real gap precludes rigorous statistical inferences about real-world outcomes from simulation results alone”.

Reproducibility. 62 rows released code that could be parsed against the paper, and 38 of the 112 rows record a disagreement of some kind between the paper and that code. All 38 released code, so the raw rate among code-releasing rows is 61 percent. That raw rate is not the finding, because the 38 are not one thing. Section V-F classifies them: 8 contradictions, 14 limitations of this survey’s own parsing, 8 components never released, 4 version skews and 4 inconsistencies internal to a paper. Only the contradictions are a finding about the work rather than about this survey, so 9 of 62 code-releasing rows, which is 15 percent, is the figure this section and Table III use. PhysHOI [143] is the clearest of the 9. It lists a non-zero object-orientation

weight for GRAB in its Table 4, and the reward function in the repository at the commit this survey fetched sets that orientation error to zero, so in that file the object is tracked in position only.

B. Physical plausibility

One quantity is specific to hands, and it is the one closed-loop policies do not record: how far the hand passes into the object it is holding. Eleven of the 96 method rows whose notes settle the question handle interpenetration at all, and every one of them measures at a reference rather than at a rollout. That is a narrower claim than it sounds, and reading it more widely would be wrong. Outside closed-loop control the quantity is a standard comparative column and has been for years: BimanGrasp [42] fails any grasp past 1.5 mm, TopoRetarget [144] reports a maximum depth and a share of frames past 2 mm against a baseline retargeter, and OakInk [174] scores a dataset split on depth, intersection volume and simulation displacement. The gap is specific to learned closed-loop control: seven of the eleven score a pose, a trajectory or a contact model before anything executes, the four that are closed-loop policies score the references they were given, and none reports the measurement for rollouts of its own trained policy. Appendix A gives the audit behind that null, which sampled 25 of the 85 rows recorded as not addressing penetration and recovered nothing.

The null result was audited. Earlier audits showed that the extraction could miss reported values, so 25 of the 85 apparently silent methods were drawn with a fixed random seed and checked again against their papers and repositories. None reports penetration depth, intersection volume, or another quantitative plausibility measure on its own rollouts. With zero recoveries, a one-sided 95 percent finite-population bound allows at most 8 hidden cases among the 85. The reported count of eleven is therefore a lower bound and nineteen an upper bound.

GraspXL [48] is the closest borderline case: human raters considered interpenetration as one component of a broader realism score. We treat that as a judgment rather than a penetration measurement. Other near misses concern self-collision, trajectory constraints, or a solver’s separation setting rather than measured hand–object overlap during a rollout. The sample, search terms, and result for every audited row are included with the corpus; Appendix A gives the full procedure.

Where in the pipeline those eleven act is the reference-versus-rollout split that Section I takes from [175]. A reference is a pose or a trajectory scored before execution, and a rollout is what the trained policy actually did. What this section supplies on that axis is a measurement method and a count, and it does not supply a threshold. The count is the eleven of 96 above, with four closed-loop policies inside it and none we found reporting a number for its own rollouts. The method is the plausibility row of Table III: maximum and mean penetration depth over the evaluation rollouts, on a dense surface sample, computed by code that never entered the reward or the termination rule. The threshold is borrowed, and Section VII-D says from where and why it does not bind. Six of the eleven are grasp synthesisers

or trajectory optimisers, namely [158], [42], [151], [36], [144] and [153], and Castro et al. [93] give a contact model rather than a controller. That leaves four closed-loop policies in the whole corpus: [176], [19], [97] and [114].

The reason only four are closed-loop is a measurement trap: a quantity a policy optimises cannot also judge it, because the policy learns the measure rather than the property the measure stands for. PhysHOI [143] documents the failure in the clean direction, its kinematic imitation reward maximised by not touching the object at all because contact perturbed the reference trajectory. DexTrack [97] shows the same trap from the other side: it defines a maximum hand-object penetration depth over all frames, applies it only to its input kinematic references, and then reports tolerance of “severe hand-object penetrations” as a result. TopoRetarget [144] is the strongest case in the corpus and still stops one step short on the same line. It constrains penetration during retargeting with a 1 mm soft tolerance and a 30 mm hard bound, and reports a maximum penetration of 1.07 mm and 0.00 percent of frames above 2 mm against 22.22 mm and 96 percent for its baseline retargeter, measured on retargeted references rather than on a rollout.

Definitions are not shared either. GRAB [177] estimates contact by proximity, because “contact cannot be directly observed”, and reports that “‘Use’ grasps have 3.25 ± 0.68 mm average penetration” without saying whether that is a maximum, a mean or a median. OakInk [174] supplies the fullest published vocabulary: penetration depth, solid intersection volume and simulation displacement. The analytic tradition scored a grasp without simulating it at all, in the wrench-space terms of [34] and [178]; neither could be obtained for this survey, and no definition here rests on their contents.

Reporting penetration. The gap is specific to learned closed-loop control rather than general, and it is a choice rather than a capability. Evaluations should report maximum and mean penetration depth over the evaluation rollouts, on a dense surface sample, computed by a measure the policy never optimized.

C. Statistical practice, and a proposed protocol

Kress-Gazit et al. [171] give the field’s reference protocol, and it prescribes process, not numbers. Write the success criteria before the run and have someone other than their author score the runs. Match initial conditions with image overlays. Interleave the policies blind within one session. Report counts rather than percentages, alongside the initial conditions and the failure modes. Use a posterior over the Bernoulli parameter instead of a point estimate. It prescribes no minimum trial count anywhere and no frequentist confidence-interval width anywhere. Its own example report uses 10 initial conditions with two runs each, 20 evaluations per policy, and its worked case shows what 20 buys. Pancake success of 15 of 18 against 11 of 17, nominally 83 against 65 percent, leaves a 0.11 posterior probability that the worse policy is actually better. At 150 of 180 against 110 of 170 the same rates separate.

The TRI LBM Team [179] supplies the missing numbers by fiat rather than derivation: 50 rollouts per task per policy per condition on hardware, 200 in simulation, blind, with

randomised policy order inside per-initial-condition bundles. It replaces confidence intervals with Beta-posterior violins and corrects its pairwise tests under Bonferroni, “unless otherwise noted”. Its own warning is the strongest sentence in this literature: “there is significant risk that many robotics papers are measuring statistical noise due to insufficient statistical power”.

The rest fall short of their own advice. RoboArena [180] runs 612 double-blind pairwise comparisons across seven institutions and 4284 rollouts, and reports no confidence intervals and no per-policy trial counts. The Colosseum [169] evaluates 235 test sets at 25 episodes each with “one training seed and one evaluation seed” per baseline and gives no intervals in simulation. AutoEval [172] runs 50 trials per policy per task and calls ± 10 percent “the natural variance of robot evaluations”, without giving the formula behind the intervals it plots.

Two papers do give usable numbers. Badithela et al. [173] pair real and simulated trials and de-biases the simulation with a rectifier, saving 20 to 25 percent of hardware trials at paired correlations of roughly 0.6 to 0.7, and nothing at all at a correlation near zero. The 25 percent figure is the better of its two reported settings, the DP case at $\rho = 0.702$. Its decision rule is exact: combining helps only when the rectifier variance is below the variance of the real evaluations. Snyder et al. [170] give the largest saving. On the LBM rubrics their sequential test on graded scores cuts simulated evaluation by about 70 percent and hardware by about 45 percent, 286 trials against a nominal 500, with per-task decisions landing in 12 to 36 paired trials. On RoboArena’s data a 30-point gap on continuous progress scores reaches significance in 18 trials, while a 20-point gap on binary success needs about 80.

A proposed protocol. Every count below is printed by `tools/make_eval_tables.py` in its derivation mode, and each axis is derived for the statistic that axis actually reports: a single rate takes a Wilson half-width, a matched comparison takes McNemar, a ratio takes the standard error of the log ratio, a correlation takes the Fisher-z interval.

Appendix E gives the full derivation; its conclusions are as follows, and every count in Table III is one of these. A single reported rate takes 100 real rollouts per cell, because 93 is where the worst-case Wilson half-width of (1) falls to 10 points and 10 points is the coarsest width at which a rate is worth printing. A comparison is a different question and a harder one. The design is paired, so the count follows McNemar and depends on the discordance rate rather than on the two rates alone, and separating 50 from 70 percent at 80 percent power takes 57 matched pairs per arm at the assumed discordance of 0.3, which is 114 rollouts against the 186 two independent arms would need. Simulated cells take 200 episodes for a 6.9-point half-width. Perturbation axes are screened at 40 per axis, which ranks the axes and cannot establish that any one of them hurt, and certified at 101 per arm. Unseen objects resample the object rather than the trial, at 20 objects and 5 trials each, and a 10-point claim about an object distribution needs about 93 objects. The transfer axis takes the 100 real trials paired to 100 simulated, and about 457 pairs to separate two correlations 0.1 apart. One hundred is a cap and not a bill, because a sequential test on a graded score reached its decision

in 12 to 36 paired trials in [170], and a cell that stops early reports a confidence sequence rather than a Wilson interval, never both. The appendix also keeps the counts an earlier draft of this section quoted and this one withdrew, since the test they were computed for was the wrong one.

D. Feasibility and limitations

The proposed protocol has not been run. In the twelve-method comparison assembled for this survey, no method supplied all the measurements needed for a complete protocol row. DeXtreme [15], for example, reports a success criterion and a confidence interval, but uses ten trials. This limits retrospective comparison; it does not imply that the reported experiments failed their own stated objectives.

What would have to be true. The protocol is expensive. For each policy and task it calls for 57 matched anchor trials and 100 unseen-object trials. Comparing two policies on three tasks therefore requires 942 real rollouts, or about 16 hours at one minute per rollout before repairs and scoring. Pairing saves roughly a fifth relative to an independent design because each pair still requires two rollouts. AutoEval [172] completed about 850 WidowX episodes in 24 hours with three interventions and regular cooling pauses; a tendon-driven hand is likely to stretch the same workload across several days.

That bill is large but not unprecedented. AutoEval [172] records that evaluating OpenVLA against its baselines took more than 2500 rollouts and more than 100 hours of human labour across three institutions, and the TRI LBM Team [179] analyzed about 1800 real rollouts across nine hardware stations. What is unprecedented is paying it for a single dexterous-hand paper, where the modal per-cell count is 10 trials and the median per-cell count 15, and where the median stated count has been 20 or 22.5 in each of the last three years.

Four things would have to change. Reviewers would have to reward 57 matched trials on one task over 20 unmatched trials on five, and nothing in the corpus suggests that is happening. The loop would have to be automated, and AutoEval [172] shows it is buildable for a gripper at one to three hours of setup, while stating that it supports binary success only and no robustness axes. Somebody would have to run the penetration measure on their own rollouts, which is a choice rather than a capability: Section IV-B shows the depth is computable from the poses and the meshes in a few lines of Warp, and IsaacGymEnvs already ships a task that does it every step. An earlier draft of this section made the engine the barrier, and that claim is withdrawn, because the released code refutes it. And the comparison would have to be sequential, because the savings in [170] are the only reason 100 is a cap rather than a cost.

The proposal has four limits. It has not been run, and its sample sizes come from power and interval calculations rather than dexterous-hand experiments. The worst-case calculation assumes a success rate of 0.5, while the paired design assumes a discordance rate of 0.3 that the literature does not report. Its perturbations come from parallel-jaw benchmarks [169], [181], where contact is simpler. Finally, the 2 mm penetration threshold follows [144]; it is a simulator convention, not a

TABLE III: The proposed evaluation protocol, and this survey’s own contribution. No cell is empty: every cell is a proposal.

axis	what is measured	minimum trials	reported with it
Task success	fraction of episodes meeting a criterion written before the run, and a graded rubric score in $[0, 1]$	100 real and 200 sim per policy, task and condition for an absolute rate at a 10-point interval; a comparison alone needs 57 matched pairs per arm	raw counts, the criterion verbatim, the rubric, the initial-condition protocol, and an interval whose kind is named
Robustness	success under each perturbation axis, and the ratio to the unperturbed anchor	40 per axis per policy to rank the axes; 101 per axis and 101 at the anchor before any claim that a named axis hurt	both absolute rates with their Wilson intervals, the axis ranges, and the ratio without an interval unless the certifying count was run
Unseen objects	mean over held-out objects of per-object success	20 objects at 5 trials to screen; 93 objects for a 10-point claim	the object list, the per-object rates, a bootstrap interval over objects, and the within-object trial count
Physical plausibility	maximum and mean hand-object penetration depth per rollout, and the fraction of frames above 2 mm	100 rollouts, simulation only	an interval over rollouts, the sample density, the threshold, the solver’s depenetration settings, and at least one rendered rollout
Sample and wall-clock cost	environment steps and wall-clock to the checkpoint that produced the headline number	3 training seeds	the range over seeds, and the statement that 3 seeds is a range and not an interval
Real-robot transfer	paired real and simulated outcomes on matched initial conditions, their correlation, and the rectifier variance against the real variance	the 100 real trials paired to 100 sim; about 457 pairs to separate two correlations 0.1 apart	the correlation with its Fisher-z interval, the variance ratio with its bootstrap interval, and which decision the count supports
Reproducibility	the released config and checkpoints against the paper’s stated objective	not a trial count	the file and line of every disagreement, or an explicit statement that none was found

physical limit. Captured human grasps in GRAB [177] average 3.25 mm under the same kind of measurement. Establishing a meaningful threshold will require measurements across hands, engines, and accepted behaviors.

The methodology literature has no dexterous hand in it. Seven corpus rows are evaluation protocols, a small denominator, and not one of them uses a dexterous hand: Badithela et al. [173] run a parallel-jaw gripper and the other six state no hand. Everything proposed above is therefore assembled from work on grippers and on whole-arm tasks. A modest first test would run the protocol in Table III once, on one in-hand reorientation task with one 16-DoF hand, and release the rollouts and per-trial outcomes for independent analysis.

VIII. CONCLUSION

Progress in dexterous manipulation now depends as much on verification as on new methods. The evidence in this survey supports seven conclusions.

- 1) Eight of the 62 method rows that released parseable code state, in a named file at a named commit, something other than the value their paper prints, and eight further rows have been withdrawn from that count since its first draft, one of them because an author replied, with two more narrowed. Table II gives each of the eight as a repository, a commit, a file and two values, and all of their authors were written to before this was posted: Section V-F says that beside the finding, with the letters in `outreach/` and the route by which a disputed case is corrected (Section V-F).
- 2) No closed-loop policy in the corpus reports interpenetration for the rollouts of its own trained policy, and every one of the eleven rows that handles penetration at all sits on the reference side of the reference-versus-rollout split (Section VII-B).

- 3) The evaluation-methodology literature this survey builds its protocol on contains no dexterous hand at all: of its seven corpus rows, one runs a parallel-jaw gripper and the other six state no hand (Section VII-D, Table III).
- 4) The generalist and vision-language-action policies that do evaluate on a multi-fingered hand run it at a median of 6 actuated degrees of freedom, against 16 across the reinforcement-learning rows that state a count, and the rows reaching the larger band on paper never say whether the hand was in the evaluation (Section V-D).
- 5) Eight hands that can be bought today or built from published designs take zero method rows between them, while the corpus’s experiments concentrate on four designs (Section III-D, Fig. 3).
- 6) Twenty-one of the 28 rows that put a learned closed-loop controller on two multi-fingered hands run one policy over a concatenated two-hand observation, a choice compared twice with opposite outcomes and ablated once (Section VI-B, Fig. 10).
- 7) Human data does not port across hands, and the map is usually unstated: 33 of the 53 method rows that use human data never say how the human motion reached the robot hand (Section V-C).

Six concern gaps in the literature. The first concerns publishing practice; the corrections made during this review are recorded beside the result in Section V-F.

Three of those claims carry a case worth remembering. A reward table in a paper is a claim about a document, not about a run, and PhysHOI [143] is the instance to keep in mind: the term its table weights at 0.1 is set to zero in the file at the commit this survey fetched, and the success criterion the paper reports is itself position-only. Contact is what separates a hand from a gripper, and the obstacle to measuring it is not the engines. IsaacGymEnvs ships a task that computes a per-

environment maximum interpenetration depth against meshes and gates the policy update on a 1 mm threshold, and Tactile Genesis [182] offers penetration depth on Genesis geometry as a sensor. The tooling sits in the field’s own benchmark repository and the number is still not reported. DexTrack [97] has the formula and points it at its inputs. And hardware and software have come apart on a narrower claim than the hand count first suggests, because 11 of the 19 hands that take no method row are neither sold nor open and take none for that reason, which is why the fifth claim is eight hands and not nineteen.

What this survey cannot establish is which method is better than which. It re-runs nothing, and Section VII argues that the published numbers do not compare. Six works are cited by metadata only, and a seventh, Ma and Dollar 2011, is on disk but unread; no claim rests on any of them. Every coverage statistic here counts what this survey’s extraction captured rather than what the literature reported. Each is a floor and not a rate, because every miss converts a reporting paper into a silent one.

Several practical changes require little additional computation. Authors can generate reward tables from the configurations used for the reported run, cite the exact revision, state the trial count and success criterion, and release per-trial outcomes. Evaluations can report penetration depth over policy rollouts as a diagnostic, without optimizing the policy against that measure. TopoRetarget [144] shows how informative the result can be. Hardware choices also deserve a broader base: the Allegro, Shadow, Inspire, and LEAP hands account for most of the published use in this corpus, while every other listed hand appears in eight method papers or fewer. Better reporting would make results easier to reproduce; broader hardware coverage would show which results survive a change of embodiment.

APPENDIX A METHOD

The survey is written from a corpus that lives on disk. No claim about a paper is made from memory. Each claim traces to a note, each note to a parsed markdown file, and each markdown file to a PDF or repository with a recorded hash or commit.

Selection. Six topic-specific bibliographies were assembled in parallel (simulators and physics, hands and vendors, reinforcement learning, imitation and human data, bimanual, benchmarks and evaluation), seeded with the canonical works in each area and extended by search up to 2026-09-17. Every arXiv identifier was checked by fetching the abstract page and matching the title, and entries that could not be checked that way are marked. The six lists were merged with deduplication on arXiv identifier and normalized title, giving 221 entries.

Seven further bibliography entries are not part of that corpus and are not counted anywhere in this survey. They are the prior audits and case studies Section I positions this survey against, they sit in a bibliography of their own as prior work from outside the corpus, and none of them carries a structured row. No source for them was parsed either, so each is quoted only from its abstract and its stated method.

Acquisition. PDFs were downloaded from arXiv or, for work without a preprint, from the publisher or vendor page recorded in the bibliography. Vendor pages for hands without any paper were fetched as HTML and converted to markdown, so that a specification quoted in this survey is quoted from a stored copy of the page and dated. Repositories were shallow-cloned and parsed, then deleted. What remains is one markdown per repository holding the README, a pruned file tree, the task and reward configuration files, and the bodies of reward and observation functions.

Parsing. PDFs were converted with pymupdf4llm, falling back to raw text extraction when the layout parse returned too little. The corpus manifest records, per entry, the source URL, page count, sha256 of the PDF and the converter used. Two consequences matter for reading this survey. Equations rendered as images do not survive conversion, so where a reward weight exists only inside a figure it is recorded as unreadable rather than guessed. Tables are flattened, so a number taken from a table is quoted with the table it came from and, where the flattening is ambiguous, the ambiguity is stated.

Notes. Each entry was read into a structured note under a fixed template: embodiment, learning method, objective, contact handling, evaluation, reproducibility, stated limitations, and quotable claims. Notes were written only from the parsed files. Where the source is silent the note says “not stated”, and nothing was inferred from the reviewer’s prior knowledge of the work. For method papers the reward or loss was quoted from the paper and, separately, from the released code, so that disagreements between the two are visible rather than smoothed over. Those disagreements turned out to be common enough to become a finding in their own right. The reading into notes was done with language-model assistance, from the parsed files only. The template’s conventions are the safeguard against what that can get wrong: a quotation rather than a paraphrase, “not stated” rather than an inference, and a named source beside every value, so a note can be checked against the file it came from. Four fields carrying a headline claim were re-read against their sources afterwards, the trial counts, the success criteria, the unseen-object counts and the penetration field, and all four audits are below; the coverage statistics are reported as floors for the same reason.

Sources that could not be obtained. Six works are cited by metadata only and no claim in this survey rests on their contents. Five are behind publisher paywalls with no author-hosted copy found on 2026-09-18: Okamura et al. 2000, Bicchi 2000 (IEEE T-RO), Piazza et al. 2019, Roa and Suarez 2015, and Butterfass et al. 2001 on DLR-Hand II. The sixth is Hwangbo et al. 2018 on RaiSim, whose hosted PDF returned no body. The freely circulating PDF often taken for Bicchi 2000 is a different work, a book chapter, and is listed separately.

Ma and Dollar 2011 is a seventh case with a different cause. The fetch that failed when its note was written succeeded on 2026-09-18, so a seven-page PDF and its hash are in the manifest, but no note has been read from it. It is cited by metadata only for that reason and not because the source is unavailable.

Three of the 221 bibliography entries carry no structured row. Two are the paywalled Bicchi 2000 and DLR-Hand II

entries. The third is [1], which was read into a note and quoted throughout but is a book chapter rather than a work with an embodiment, a method or a result to record in a row.

The reference list of this edition is shorter than the corpus, and the two numbers are different quantities. It prints 208 entries: the 201 corpus entries that some sentence, table or figure of this paper cites, plus the 7 prior-work entries from outside the corpus. The other 20 corpus entries carry a row and a note and are counted in every statistic here, but no passage in the paper names them, so they have nothing to be cited from and do not appear in the list. A reader counting the reference list should get 208, and a reader counting the corpus should get 221.

What this method cannot do. Mention counts over the corpus are counts of mentions, not of use: a related-work sentence counts the same as an experiment. Vendor specifications are manufacturer claims and are labeled as such throughout, and where a page has since gone offline the note says so. The corpus is large but not exhaustive, and selection by search favours work that is indexed, in English, and posted as a preprint. It also favours recent work. Of the 221 bibliography entries, 138 are dated 2024 or later and 16 predate 2018, so this is a corpus of the learned era and any claim here about a trend over time is a claim about 2022 onward.

Every coverage statistic in this survey measures what this survey’s extraction captured, not what the literature reported. A structured row holds a scalar. A paper that reports a per-task count, a rubric, or a total spread across several tables produces a null, and a null is then counted as silence. The bias runs one way. Every miss converts a reporting paper into a silent one, so the field is made to look worse at reporting than it is. The size of the effect was measured on the statistic the survey leads with. Of the 34 method rows that had a real robot and no recorded trial count, 15 carried a count in plain text in their own note, dropped because the paper reports it per task and the field takes a single integer. Those 15 have since been re-extracted, which moved the stated-trial-count row from 55 to 70. The success criterion and the unseen-object count were audited the same way, rising from 79 to 98 and from 32 to 39. What those 19 recovered criteria say is mostly not a threshold: eleven score by rubric or staged partial credit, five judge binary completion against a task description by eye, two defer to a benchmark’s own definition, and exactly one, [24], states a verbatim numeric threshold. The audit counted only cases where the note itself carried the number, so a count the note also missed is still uncounted, and 19 trial counts, 8 unseen-object evaluations whose object count the note never gives, and 4 criteria remain genuinely unsettled. Fifteen of the recovered trial counts are named in the rows: π_0 [22] at ten trials per task, RDT-1B [154] at 139 across seven tasks, UMI [157] at 260, $\pi_{0.6}^*$ [24] at 750, Gemini Robotics [147] at twenty per task, and ten more, which is 44 percent of the audited nulls and moved the “never says” figure from 34 of 89 down to 19. Read every coverage statistic in this survey as a floor rather than as a rate, and read the bars in Fig. 13 the same way.

The penetration field, audited the same way. The same mechanism reaches the penetration, code-release and failure-

mode fields. The penetration field has since been audited by hand, because this survey’s second finding is a null over it and a null is worth what the search behind it is worth. The population is the 85 method rows whose contact-handling field records that the work does not address penetration. It excludes the 16 nulls, which are already counted as unsettled rather than as silence, so recovering one would not move the finding. Twenty-five of the 85, 29 percent, were drawn with `random.Random(20260919).sample` over the sorted population, so the draw is fixed and can be redrawn. Each sampled row was read again in `papers/md/`, in its OCR recovery where one exists, and in `code/md/` where a repository was parsed — and not in `papers/notes/`, because the note is the artifact under suspicion: a field is wrong exactly when the source says something the note did not carry. A regular expression over the whole source collected every occurrence of penetration, interpenetration, intersection, intersection volume, solid intersection, simulation displacement, contact consistency, physical plausibility, signed distance and contact depth, and of the words that are mistaken for them, and every hit was read in its context. A row counted as a recovery if its own source reported a measurement of any of those quantities on that method’s own rollouts. A statement that penetration occurs, a citation whose title says physically plausible, a collision-avoidance constraint on a reference trajectory and a solver setting did not count, and the three recurring near misses of that kind are listed in Section VII-B. Recoveries: none of the 25, against the 20 to 45 percent the three earlier audits recovered on the fields above. The one-sided 95 percent bound, computed hypergeometrically over the finite population, is therefore 8 of the 85, and Section VII-B states it beside the claim and names the one borderline case that a reader might count differently. Why this field held where the others did not is visible in the per-row results: the recovered trial counts and criteria were numbers present in the source and dropped by a field shaped to hold a scalar, whereas a penetration number is absent from the source altogether. The earlier audits measured a defect in this schema; this one looked for an absence in the literature. `reviews/penetration_audit.md` holds the seed, the sample and a result per row beside the sentence it rests on, and `tools/audit_penetration.py` redraws the sample and re-runs the sweep. The code-release and failure-mode fields have still not been audited this way, and their counts stay floors.

Where the tabulation lives. Most of this survey’s tabulation is not printed in this paper. The 112 method rows, the 21 reward-term objectives scored against their released code inside a 77-row extraction whose ten columns hold 253 cells no source stated, the 30 teleoperation systems, the 33 hand specifications with the source sentence behind each one, the 15 contact-model and solver descriptions verbatim, and the cell-by-cell mapping behind the predecessor-survey comparison: all of it is reference material, consulted a row at a time and never read through. Printed, it ran to more pages than the argument it supports. It is in the repository instead.

Read it there in one of three forms. `corpus/rows/` holds one JSON record per work, with every field at full length and no cell cut; this is the form to query, and it is the form every

number in this paper is computed from at build time. `paper/tables/` holds the same content as `markdown tables`, one file per table, for reading rather than querying. `papers/notes/` holds the structured note each row was read from, with the quotation that supports each field, so that a cell can be traced to a sentence in a parsed source. The generators are in `tools/`: every table in this paper and in the repository is written by a script there from those records, so a reader who disagrees with a classification can change it and rebuild rather than argue with a PDF.

The corpus is not deposited yet, and this paper prints no link to it for that reason. It is available from the author at the address in the author block, and the archived release will carry a persistent identifier, which will be the copy to cite for any number quoted here. A survey that argues for checkable artifacts should point at its artifact rather than reprint it, and a table a reader cannot recompute is worth less than the record it was compressed from.

Three tables are exceptions and are printed, because the prose reads individual cells out of them and does not stand up without them: the two hand tables and the engine table, in Appendix B. The proposed evaluation protocol appears in the body alongside its justification. Table VII is the comparison to the fourteen predecessor surveys, which is this survey's positioning and is therefore set where a reader meets it rather than in the repository. It prints one of three levels per cell; Appendix D is the prose those levels coarsen, and the cell-by-cell mapping remains in the reproducibility materials.

Reproduced figures. One figure in this edition is not this survey's own artwork. It is a render reproduced from the paper it discusses, rendered from the stored PDF in the corpus, and it states its source in its own caption, keyed to the corpus entry it came from and to the page it was taken from. An earlier draft reproduced eighteen. A plate is kept only where it carries something a drawing cannot, and only where this paper has the right to print it: a contact defect invisible at the scale of a grasp and unmistakable at the scale of the contact, on a source that is CC BY 4.0. The other seventeen were redrawn in the paper's own style or dropped, and what each one was standing in for is said in the comment at the head of the figure that replaced it, or in the paragraph that absorbed it. Two of those seventeen went last and for a reason worth stating, because it is a limit on what this paper shows rather than a tidying of its figures. Both would have needed an author's permission, since arXiv's non-exclusive license to distribute grants a third party nothing, and this paper is posted without asking anyone for anything. One of them argued from geometry and was redrawn, as Fig. 8. The other photographed five research hands to scale beside a human hand and a ruler, and it was dropped rather than redrawn: the honest redraw is a to-scale outline annotated in millimetres, and no source in the corpus states a hand's length or its width, so every envelope in such a drawing would have been invented. Section III now states that absence instead of illustrating around it. The one surviving plate is cleared: its source carries CC BY 4.0, read directly from the license page on the date the register records, and its caption prints the attribution the license asks for, naming the creator, the work, the license with a link and the change made to the plate. No

permission request and no author email is outstanding, and nothing in this paper waits on a reply. The register that records, per figure, the license found, the URL it was read from and the position it closed at, is `tex/PERMISSIONS.md`. The build does not enforce that register.

APPENDIX B THE HARDWARE AND ENGINE TABLES

Sections III and IV argue over individual cells of three tables: which column is emptiest, which cell is blank and why, which figure rests on a press release. Those three are set here rather than in the body, because a reader consults them rather than reading them through. Every other tabulation in this survey is in the repository, and Appendix A says where.

Nothing in these tables is a measurement by this survey. Every specification is a manufacturer claim, a figure in a paper, or a figure in a press article. Table IV and Table V split the 33 hand rows by whether the hand can be bought or built, and Table VI carries the 15 engines. The controlled vocabularies in all three compress a sentence the corpus stores in full: `corpus/rows/` holds the actuation, tactile, force-kind, contact-model and solver text at its own length, with the source's own punctuation and wording, and `papers/notes/` holds the quotation each was read from.

Each caption states how many of its own cells no source filled. Two conventions behind those counts are worth stating once. A blank price is a hand with no public list price rather than a free hand; and a hand with no entry in the manifest was recorded from a page that could not be stored, so its row should be read as a claim with no retrievable source.

Table IV holds the hands that can be had: sold, open-source, or with no release status stated. A row whose joint count no source states sorts last rather than being ranked by a number nobody stated. Its empty-cell count covers the eight specification columns, DoF through price; the hand, maker, open-hardware, status and usage columns are left out of it because they are never blank and counting them would dilute the finding. *force N* is not a ranking: read *force kind* first, because pull-out resistance, pinch force, a fingertip normal force under an indenter and an unlabelled vendor number are different measurements. *uses* counts the method rows whose own experiments run on that hand.

Table V carries the columns of Table IV plus the date of the claim and the class of artefact it was read off, over the same eight specification columns. Its empty share is far higher, and that is the finding of the table rather than a defect of it: an unreleased hand is announced without a specification. *evidence* is *datasheet* for a paper, *datasheet* or *vendor specification* page, *video* for a demonstration reel, *press* for a press release or a third-party tracker. *claim date* prints *undated* where the source page carries no date of its own; a fetch date is not a claim date.

Table VI prints, for each engine, the contact formulation it names, whether it exposes interpenetration depth and how many first-party hand models it ships; its empty-cell count is over the seven property columns. *contact model* and *solver* are controlled vocabularies: *soft* for a compliant or regularised

contact, *convex relaxation* for a convexified formulation, *complementarity* for an LCP, *hard NCP* for a nonlinear complementarity problem, *impulse* for a velocity-level update, and *rigid, model unstated* where the source describes collision geometry but names no contact law. The sentence each was read from runs to several hundred characters; *corpus/rows/* carries each one verbatim, because Section IV’s argument turns on its wording. *penetration exposed* is whether the engine makes interpenetration depth readable by a user’s code without patching it. *hands* counts first-party hand models in the parsed snapshot, and they are: *Brax* synthetic 4-fingered claw hand; *Genesis* Shadow Hand; *Isaac Gym* Shadow Dexterous Hand, *Allegro Hand*, *TriFinger*; *Isaac Lab* KUKA *Allegro hand*, *ShadowHand*, *AbilityHand*, *Fourier GR1-T2 dex hands*, *Unitree Inspire hand*, *Unitree trihand*; *ManiSkill3* *Allegro Hand*, *Ability Hand*, *Inspire Hand*, *Delto 3-finger hand*, *TriFinger*; *Newton Allegro Hand*; *Orbit Allegro hand*.

APPENDIX C

PAPER AGAINST RELEASED CODE, IN FULL

Each entry below is the disagreement text stored in the row, unedited, and the class beside it is the one Section V-F and Section VIII count. A *contradiction* is a paper stating one value where the shipped code demonstrably states another. A *parse-limitation* is a comparison withdrawn because this survey’s own parse could not settle it. *Code-absent* means the described component is not in the released repository, *version-skew* that the repository is a later generation than the paper, and *internal-inconsistency* that the paper disagrees with itself with no code implicated. Where a comparison was narrowed or withdrawn under review, the review note is printed beneath it. Every *contradiction* entry also carries an *Artifact* line: the repository, the commit *corpus/code_manifest.json* records, the file inside it and where in that file the value sits, so the entry can be settled without asking anyone. All of those authors were written to before this survey was posted. Section V-F says so beside the finding, sets out what a fetched commit can and cannot show, and states the route by which a disputed entry is corrected; the letters are in *outreach/* in the corpus, which is available from the author at the address in the author block. The reward-term extraction these rows were read out of is in the repository rather than here, with the rest of the tabulation Appendix A describes.

contradiction, 9 rows.

- PDDM [109] (high). Table 2 states obs-dim 46 for In-hand Reorientation while the released *cube_env.py* code sums to 39; the Baoding reward code includes an extra $-10 \times \text{wrist_too_high}$ term absent from Table 2’s printed formula.
Artifact: `google-research/pddm` at `06b88cdbaf`, `pddm/envs/cube/cube_env.py` (`_get_obs`).
- DexPoint [38] (high). The released code’s reward adds several terms absent from the paper’s four-term Eq. 5 (a lift-threshold bonus, a target-distance term, a rotation bonus, and an IK controller-tracking penalty) and reshapes the reach/lift terms into inverse-distance and clipped forms rather than the paper’s plain distance/height-difference

formulas.

Artifact: `yzqin/dexpoint-release`
at `17f1e238bb`, `dexpoint/`
`env/rl_env/relocate_env.py`
(`AllegroRelocateREnv.get_reward`).

- DeXtreme [15] (high). Action Delta Penalty weight is -0.25 in paper Table 2 but -0.2 in the ADR yaml and -0.01 in the ManualDR yaml; the Joint Velocity Penalty in code normalises velocity by $(\text{max_velocity} - \text{vel_tolerance})$ unlike the paper’s stated formula; code has a `timeout_rew` term absent from the paper’s reward table; Appendix Table 12 states critic learning rate $5e-4$ and KL threshold 0.16, vs body text/code values of $5e-5$ and 0.016.

Artifact: `isaac-sim/IsaacGymEnvs`
at `aed298638`, `isaacgymenvs/cfg/`
`task/AllegroHandDextremeADR.yaml`
(`actionDeltaPenaltyScale`).

- Visual Dexterity [17] (high). Table S1 states 32000 teacher training environments, but the released config sets `alg.num_envs` to 8000 (parent config 16384); `fallDistance` differs across two shipped configs (0.24 vs 0.15, only the latter matching Table S1’s threshold); the paper’s Eq 8 penultimate-joint penalty ($c7=-2$) does not appear anywhere in the released reward code; the config carries a dead `distRewardScale=-10.0` key never used in `compute_reward`; and the paper’s table-friction lower bound (0.05) differs by a factor of 10 from the code’s randomized lower bound (0.005).

Artifact: `Improbable-AI/dexenv` at `ad9634e9d2`,
`dexenv/conf/dclaw.yaml` (`alg.num_envs`).

- DexPBT [111] (high). Paper presents the reward as 4 mutually exclusive stage terms (`r_reach`, `r_pick`, `r_targ`, `-r_vel`), but code’s `compute_kuka_reward` sums 8 named components, one of which (`hand_delta_penalty`) is multiplied by 0 and disabled; there is no single `r_vel` term in code, instead separate `kuka/allegro` action penalties whose exact formula is not shown. Also, the paper reports zero experiments with domain randomization, yet the shipped `AllegroKuka.yaml` already carries a fully specified DR schedule (disabled via `randomize: False`).

Artifact: `NVIDIA-Omniverse/IsaacGymEnvs` at
`aed298638`, `isaacgymenvs/tasks/allegro_`
`kuka/allegro_kuka_base.py` (`compute_`
`kuka_reward`).

Review: R3 adversarial review: the disabled-randomisation half is withdrawn, since the note finds it consistent with the paper; the zeroed reward term stands

- PhysHOI [143] (high). The released `compute_humanoid_reward` sets the body position-velocity error and the object rotation and rotation-velocity errors to `zeros_like`, unconditionally, with the computation that would produce them commented out on the same lines, while Table 4 lists nonzero $\lambda^{or}=0.1/\lambda^{orv}=0.01$ weights for GRAB: in that file the object’s orientation error is the constant zero and those weights cannot change the reward.

Artifact: `wyhuai/PhysHOI` at `6095c605e2`,
`physhoi/env/tasks/physhoi.py` (`compute_`
`humanoid_reward`).

TABLE IV: Hands that can be obtained, sorted by joint count. 68 of 160 specification cells are empty.

hand	maker	DoF	act. DoF	actuation	mass g	force N	force kind	tactile	price USD	open HW	status	uses
Shadow Dexterous Hand [52]	Shadow Robot Co.	24	20	tendon	4 300	–	–	optional	–	no	sold	21
Proception ProHand 1.0 [87]	Proception Inc	22	–	tendon	–	–	–	unstated	–	no	sold	0
SharpaWave 22-DoF [81]	Sharpa Robotics	22	22	–	1 300	20 / 12	grip	unstated	–	no	sold	6
BiDexHand [61]	Northwestern Univ.	21	16	tendon	–	2.14	fingertip normal	none	–	yes	open-source	0
Ruka-v2 [59]	NYU and NYU Shanghai	20	16	tendon	–	–	–	optional	1 500	yes	open-source	0
TESOLLO DG-5F [86]	Tesollo Inc.	20	20	direct drive	1 763	–	–	none	–	no	sold	0
Unitree Dex5-1 [85]	Unitree Robotics	20	16	tendon	1 100	10	fingertip normal	array	–	no	sold	0
Wuji Hand [82]	Wuji Tech.	20	20	direct drive	–	–	–	fingertip	–	no	sold	2
ORCA [62]	ETH Zurich	17	17	tendon	1 200	–	–	fingertip	–	yes	open-source	0
Allegro Hand V4 [63]	Wonik Robotics Co. Ltd.; earlier versions...	16	16	–	–	–	–	none	–	no	sold	35
LEAP Hand [60]	CMU	16	16	direct drive	595	19.5	pull-out	none	2 000	yes	open-source	12
RUKA [64]	NYU	15	11	tendon	–	2.74	pinch	none	1 300	yes	open-source	0
Inspire Robots RH56DFX [53]	Beijing Inspire Robots Tech. Co., Ltd.	12	6	–	540	10	vendor, no protocol	none	–	no	sold	19
ROBOTERA XHAND1 [183]	ROBOTERA	12	12	geared	1 100	–	–	unstated	14 000	no	sold	8
DEX-EE [184]	Shadow Robot Co., in collaboration with...	12	–	–	4 100	–	–	unstated	–	no	sold	1
BrainCo Revo 2 [185]	BrainCo Inc.	11	6	–	383	15	pinch	unstated	–	no	sold	1
AgiBot OmniHand 2025 [186]	–	–	–	–	–	–	–	–	–	–	unstated	1
DexHand [54]	IoT Design Shop	–	–	tendon	–	–	–	none	300	yes	open-source	0
LinkerBot Linker Hand [187]	LinkerBot	–	–	–	–	–	–	unstated	–	no	sold	1
PSYONIC Ability Hand [188]	PSYONIC	–	6	linkage	–	–	–	unstated	–	no	sold	7

TABLE V: Hands announced, prototyped or held internally, which cannot be obtained. 60 of 104 specification cells are empty.

hand	maker	DoF	act. DoF	actuation	mass g	force N	force kind	tactile	price USD	open HW	status	uses	claim date	evidence
Clone Robotics [79]	Clone Robotics	27	–	fluidic muscle	<910	–	–	array	–	no	internal-only	0	2025-11-24	video
1X NEO [77]	1X Tech.	25	25	tendon	–	45	vendor, no protocol	unstated	–	no	announced	0	2026-07-09	datasheet
LEAP Hand V2 [189]	CMU	21	17	tendon	–	–	–	none	3 000	yes	announced	2	undated	datasheet
Sanctuary AI Phoenix [190]	Sanctuary AI	21	–	fluidic muscle	–	–	–	unstated	–	no	internal-only	0	2024-12-17	press
ILDA hand [57]	Korea Univ.	20	15	linkage	1 100	34	fingertip normal	array	–	no	prototype	0	2021-11-05	datasheet
Pisa/IIT SoftHand [56]	Centro E. Piaggio, Univ. Pisa and IIT	19	1	tendon	–	–	–	none	–	no	prototype	0	2014	datasheet
Faive Hand [58]	ETH Zurich	16	11	tendon	1 100	–	–	none	–	–	prototype	1	2023	datasheet
Boston Dynamics [191]	Boston Dynamics	–	–	geared	–	–	–	unstated	–	no	prototype	0	2026-01-05	press
Daxo Robotics Muscle [55]	Daxo Robotics	–	–	tendon	750	–	–	–	–	–	prototype	0	2026-09-16	press
Figure 03 [80]	Figure AI	–	–	–	–	–	–	unstated	–	no	prototype	0	undated	press
PaXini DexH13 [192]	PaXini Tech	–	–	–	–	–	–	array	–	–	announced	0	2026-01-07	press
Tesla Optimus [78]	Tesla	–	–	tendon	–	–	–	unstated	–	no	prototype	0	2023-12-13	video
Xiaomi [193]	Xiaomi	–	–	tendon	–	–	–	unstated	–	no	prototype	0	2026-03-30	press

TABLE VI: 15 simulators and physics engines, by contact formulation and penetration exposure. 40 of 105 property cells are empty.

engine	contact model	solver	diff.	GPU	dt (s)	penetration exposed	hands
Brax [194]	convex relaxation	explicit/spring	yes	yes	–	no	1
ComFree-Sim [96]	complementarity-free	closed form	–	yes	0.002	yes	–
Dojo [95]	hard NCP	interior point	yes	no	–	yes	–
Genesis [100]	soft	Newton	yes	yes	–	–	1
Isaac Gym [92]	convex relaxation	PGS	no	yes	0.008333	no	3
Isaac Lab [98]	rigid, model unstated	Gauss-Seidel	no	yes	–	no	4
ManiSkill3 [104]	convex relaxation	PGS	–	yes	–	–	5
MuJoCo [102]	complementarity	Newton	no	no	–	–	–
MuJoCo convex [94]	complementarity-free	closed form	no	no	0.01	–	–
MuJoCo Warp [99]	soft	PGS	no	yes	–	–	–
Newton [101]	soft	primal-dual	–	yes	–	–	1
Orbit [103]	convex relaxation	PGS	no	yes	–	no	1
PyBullet [195]	–	–	–	yes	–	–	–
RaiSim [196]	–	–	–	–	–	–	–
SAPIEN [197]	convex relaxation	PGS	–	no	–	–	–

- UniDexGrasp [153] (high). The paper describes a four-term weighted reward ($r_{\text{goal}} + r_{\text{reach}} + r_{\text{lift}} + r_{\text{move}}$ via Table 7’s omega weights) but the released compute_hand_reward implements a different threshold-gated torch.where cascade with distinct hardcoded coefficients that do not map one-to-one onto the paper’s weights.

Artifact: PKU-EPIC/UniDexGrasp at 36c9bfcf7c, dexgrasp_policy/dexgrasp/tasks/shadow_hand_grasp.py (compute_hand_reward, goal_cond branch).

- PenSpin [73] (medium). The appendix states a randomised disturbance force and the released task config sets its scale to zero. A second half of the original comparison, that the released code disables the paper’s tactile observation channel, is withdrawn: the config read has 96 observation dimensions and enable_tactile: False, consistent with the proprioception-only student policy rather than the tactile-and-point-cloud oracle, and the paper never claims the student has tactile input. The code’s reward scale-key names (e.g. rotate_reward_scale, pencil_z_dist_penalty_scale) also do not 1:1 name-match the paper’s Table 4 weight list, though matched values agree.

Artifact: HaozhiQi/penspin at 5035c52dc9, configs/task/AllegroHandHora.yaml (forceScale).

Review: Narrowed while the letter to its authors was being drafted, before that letter went out on 19 September 2026. The tactile-channel half is withdrawn as a plausible reading of which pipeline stage the config belongs to; the disturbance-force half stands. Held at medium confidence pending a direct code read.

- PianoMime [32] (high). Paper’s Table 3 states 2 weighted reward terms (Key Press 2/3, Mimic 1/3), but the released code sums roughly 5 unweighted terms (key press doubled, sustain, energy and fingering hardcoded to return 0, forearm-collision) plus a separately-added mimic wrapper term.

Artifact: sNiper-Qian/pianomime at c4abefac8d, single_task/piano_with_

shadow_hands_res.py (_set_rewards).

internal-inconsistency, 4 rows.

- ALOHA [128] (high). Algorithm 1 pseudocode states $L_{\text{reconst}} = \text{MSE}$, but Sec.IV.C’s prose explicitly states L1 loss is used instead; the algorithm box and the implementation text disagree; code/md captures only function signatures, not bodies, for policy.py’s loss implementation
- 3D Diffusion Policy [130] (medium). the paper’s prose states the network predicts the noise added to the data, but the shipped default config trains with prediction_type: sample (predicting the denoised action a^0 directly), not epsilon; the paper only qualifies this later in the same section (Fig. 7 ablates both).
- OmniH2O [28] (medium). stumble weight -0.00125 (paper) vs -1250 (code); max-feet-height sign/magnitude differ (+1000 paper vs -2500 code, a penalty not a bonus); paper’s $\exp(-c*///l)$ form vs code’s $\exp(-\text{err}^2/\text{sigma})$; curriculum level-down threshold 40 (paper) vs 50 (code). *Review:* Withdrawn as a contradiction at the point of drafting the letter to its authors, which was never sent. Four of the five weight comparisons match the paper’s table to the digit once a systematic $\times 1.25$ curriculum factor is accounted for, and only the stumble weight differs, by a factor of about a million, which is a typo signature in the paper’s own table rather than evidence of a different trained objective. The hands in this work are also driven open-loop from VR pose, outside the policy and outside the reward, making it a poor fit for a dexterous-manipulation reward census regardless.
- DexMachina [19] (low). paper describes a plain weighted sum $\text{sum } \lambda_{\text{task}} * r_{\text{task}} + \lambda_{\text{imi}} * r_{\text{imi}} + \lambda_{\text{bc}} * r_{\text{bc}} + \lambda_{\text{con}} * r_{\text{con}}$ with unspecified weights; code implements a multiplicative task term with per-component beta decay, an unmentioned 0.1 force-penalty term, and curriculum-driven decay of auxiliary weights not described as such in the paper *Review:* R3 adversarial review: the multiplicative form credited to the code is printed in the paper itself

version-skew, 4 rows.

- Hora [76] (high). The parsed code commit’s joint-noise range, disabled default disturbance force, and default cube object type differ from the paper’s stated values; the code README itself says to use tag v0.0.1, not the parsed commit, to reproduce paper numbers.
- π_0 [22] (high). The released repo (openpi, commit 215abfb2) has evolved past this paper: it also documents pi0-FAST (autoregressive) and pi0.5 checkpoints/configs not described in this paper, which describes only the flow-matching pi0 and the non-VLM pi0-small ablation.
- GR00T N1.6 [29] (high). code/md is the current Isaac-GR00T main branch (N1.7 generation, commit 51d4c89f), not the N1.6 checkpoint; the repo names a separate n1d6 branch for N1.6 that was not fetched. The page states N1.6’s backbone is “an internal NVIDIA Cosmos-2B VLM variant,” but the repo’s own changelog states the Eagle backbone (nvidia/Eagle-Block2A-2B-v2) was used through N1.6 and only replaced by Cosmos-Reason2-2B in N1.7 – whether the page’s Cosmos-2B variant is a third distinct model or the page is describing N1.7 under an N1.6 headline is not resolvable from what was fetched.
Review: The page attributes a Cosmos backbone to N1.6 while the repository changelog says N1.6 used Eagle and N1.7 made the switch, so the repository is a later generation than the page describes.
- GR00T N1 [148] (high). The parsed code repo (Isaac-GR00T, commit 51d4c89f) is a later N1.7 generation, not the paper’s GR00T-N1-2B: VLM backbone changed from the paper’s Eagle-2 to Cosmos-Reason2-2B (via Qwen3-VL); action horizon expanded from the paper’s stated $H=16$ to 40; state/action dimensions expanded from 29 to 132; DiT layers changed from 32 to 16; embodiment scope widened to include Unitree G1, AgiBot G1, and a ‘YAM’ arm not described in the paper.

code-absent, 8 rows.

- AsymDex [41] (medium). Code contains a GraspAndPlace-task reward variant (AllegroHandDualArmGraspAndPlaceAsymDex.py) with exponential hand-distance and block-alignment terms not one of the four BiDexHands tasks described in the paper body.
- DexMimicGen [46] (high). The environment code includes an unused ‘reward_shaping’ branch (if self.reward_shaping: pass) that is never described or used in the paper text; only the sparse binary completion reward is used, as the paper states.
- DrEureka [118] (high). The released DrEureka repo (code/md) contains only forward_locomotion/ and globe_walking/ trees; the LEAP-hand cube-rotation training/deployment code and the quoted Prompt-13 reward are the LEAP-hand authors’ own code, not included in the released repo, so the cube-rotation reward cannot be cross-checked against a code file.
- Open-TeleVision [121] (low). The paper’s prose states 25k training iterations, lr 5e-5, batch size 45, but the code repo’s example training command instead shows 50000 epochs and an explicit kl_weight=10 not mentioned anywhere in the paper text.

Review: R3 adversarial review: a behavior-cloning paper with no reward; it compared paper iterations to a README example’s epochs

- Being-H0 [198] (high). The actual downstream action-head module referenced in §4.3 (learnable query tokens, proprioceptive projector f_p , regression head f_r) is not present in the parsed code; the repo’s own TODO list marks ‘Training code and scripts’ as unreleased, so the action-chunk-length=16 value and exact proprioception-vector fields come only from inference/eval CLI examples, not a verifiable implementation.
- $\pi_{0.5}$ [23] (high). The paper describes a hybrid discrete-FAST-token-pretrain-then-flow-matching-post-train recipe (Eq. 1, α : 0 to 10), but the released repo states plainly it ‘currently only support[s] the flow matching head for both pi0.5 training and inference.’
- $\pi_{0.6}^*$ [24] (high). The general openpi framework (pi0/pi0.5/FAST, flow-matching action head, ALOHA/DROID/LIBERO adapters) is released at commit 215abfb2, but the RECAP-specific pieces described in the paper – the advantage-conditioning input, the distributional value function, the RECAP training loop (Algorithm 1), and the espresso/laundry/box-assembly checkpoints and datasets – are not present in the parsed repo.
- Being-H0.5 [199] (high). Released code (commit e12ac44f) exposes only benchmark/inference/config scaffolding (policy.py, beingh_policy.py, dataset-transform configs). no MoT/MoF model definition, no rectified-flow action-expert source, and no config for any dexterous-hand embodiment; the only post-train configs present (libero_robotcasa.yaml, libero_all.yaml, robotcasa_human.yaml, so101_example.yaml) are all gripper-only, so the dexterous-hand claims cannot be cross-checked against released code.

parse-limitation, 14 rows.

- ArtiGrasp [160] (low). paper’s two regulariser weights ($w_{rh}=0.5$, $w_{ro}=0.2$) correspond to five separate velocity penalties in code (-0.5,-0.2,-0.5,-0.5,-0.3); the fingertip weight 12.0 and lambda 5.0 in Table 6 do not appear in the yaml; reward weights differ between curriculum phase-1 and phase-2 configs though Table 6 reports only one set
Review: R3 adversarial review: the reward body is C++ and absent from the parse; the weight split is the paper’s documented curriculum
- Diffusion Policy [129] (high). the paper’s core loss equations (DDPM training loss, EBM/InfoNCE, score-matching) are unrecovered images in the parsed text, and code/md is a signature-only API map, so compute_loss’s body could not be verified from code; only the exact shipped noise-scheduler config (DDPMScheduler, 100 train steps) is confirmed.
- Eureka [16] (low). The released repo’s default eureka/cfg/config.yaml ships iteration:1, sample:3 (a fast-test default), not the paper’s reported experimental configuration of $N=5$ iterations $\times$ $K=16$ samples $\times$ 5 independent

runs, and LLM-generated reward code itself is never checked into the repo (written at runtime to a gitignored outputs/ directory).

Review: R3 adversarial review: the repo README documents the paper’s defaults; the cited config block is a fast-test preset

- UniDexGrasp++ [13] (low). The paper’s four named reward weights (0.5, 0.1, 2, 10) numerically match literals in the released compute_hand_reward, but the code never labels them with the paper’s variable names, and the paper’s own display equations were dropped from the PDF, so the correspondence is inferred rather than confirmed by either source.
- BiDexHD [161] (high). Table 5 weights (w_r , w_t , w_l , w_4 , $\lambda_w=0.12$, $\lambda_{ft}=0.48$) cannot be matched to code reward lines because the code md truncates the reward function before stage-2/final sum; λ_w and λ_{ft} numerically match the code’s grasp-condition thresholds (0.12, 0.12x4 fingers) rather than confirmed reward weights
- GraspXL [48] (low). The code splits the paper’s single $r_{reg} = -w_h/u_h/\ell^2 - w_o/u_o/\ell^2$ into four separately-weighted terms (wrist vel, wrist qvel, object vel, object qvel), and the floating-phase object-velocity coefficient (-1.5) does not match the paper’s stated $w_o=0.1$, so the regularization weighting cannot be reconciled 1:1 with Table 9.
Review: R3 adversarial review: the environment source is not in the parse and the note says the reward cannot be verified
- HumanPlus [27] (medium). code implements a superset of legged_gym reward terms (lin_vel_z, ang_vel_xy, orientation, base_height, torques, dof_vel, dof_acc, action_rate, collision, termination, dof/torque limits, tracking_lin/ang_vel, feet_air_time, stumble, stand_still, feet_contact_forces, target_jt) beyond the 8 terms documented in the paper’s Table 1
- Omnigrasp [12] (medium). The code’s compute_pregrasp_reward_time hard-codes w_{pos} , $w_{rot} = 0.9, 0.1$, silently overriding the rwd_specs weight arguments passed into the function; separately, the paper’s pre-grasp reward weights (w_{hp} , w_{hr}) are not resolvable from either the paper text or Appendix Table 7.
- ResDex [152] (medium). The base-policy and residual-policy task files implement different variants of compute_hand_reward (whether finger/hand-distance terms are included alongside goal_hand_rew/hand_up/bonus differs), a difference the paper’s Sec 4.1/4.4 text does not describe, and the released configs’ env count (4096 default) and iteration counts (2500/5000) do not match the paper’s stated 11,000 envs and 5,000/20,000 iterations.
- DexGraspVLA [11] (high). Hardware-interfacing code is explicitly withheld (“Due to intellectual property constraints, we are unable to open-source the hardware-related code”); only function signatures for the controller loss (compute_loss, noise_assignment) are present in code/md, not their bodies, so the MSE/Immiscible-Diffusion loss cannot be verified against code.

- DexTrack [97] (low). The paper gives one canonical reward-weight table (Table 3), but the released repo’s task-config YAMLS contain several divergent, unreconciled reward-coefficient sets across AllegroHandTracking/...TrackingGeneralist/other variants, and it is not identifiable from the parsed source which config produced the paper’s headline Table 1 numbers.
- DexVLA [49] (high). code/md only carries class/function signatures, not bodies, for the diffusion head (modeling_scaledp.py) and no config file with the α weight was captured; the paper’s diffusion loss and denoising process cannot be verified against code bodies. only the multi-head ScaledDP signature and a DDIM fp32 patch are available, so none of the paper’s tables can be reproduced from what was parsed.
- ManipTrans [18] (low). paper’s PPO learning rate ($5e-4$) and env count (4096) differ from the shipped README/yaml defaults ($2e-4$, 8192)
Review: R3 adversarial review: the cited values are a README override and an unused fallback; the repo’s own config matches the paper

APPENDIX D

THE EXISTING SURVEYS, AND WHAT EACH COVERS

Fourteen corpus entries are themselves surveys or engine-comparison studies. Table VII in Section I sets them on one set of columns, three levels to a cell, recording what each work covers and not how well. This appendix is the prose behind those levels: each glyph in that table coarsens one of the sentences below. Section I states what this survey adds over them and does not repeat it here.

The survey of An et al. [2] is the closest in subject, covering imitation learning for multi-fingered hands by learning family, end-effector class and demonstration source. It gives reinforcement learning no taxonomy, treats bimanual work as a single “multi-agent” subsection, compares no simulator, and defines no evaluation metric. The survey of Welte et al. [200] finds only seven dexterous works that use interactive imitation learning and carries a fifteen-hand commercial table, with no bimanual section, no benchmark table and no contact modeling. The survey of Zhao et al. [201] supplies the standard sim-to-real split and predates GPU-parallel simulation. The survey of Firoozi et al. [202] has zero occurrences of bimanual, tactile or in-hand.

The survey of Bai et al. [37] spans all of manipulation across 212 pages. Its Sec. 4.3 on dexterous manipulation runs about 720 words, the third longest of its ten task subsections behind grasping and quadrupedal manipulation, which is a real treatment and not a passing mention. The difference is elsewhere. Its Sec. 1.2 lists dexterous manipulation among the topics that “existing surveys” cover from “narrower perspectives” and defers it to two of them.

The survey of Zhao et al. [175] is the most recent hand-centred one, with a 29-hand anatomy table and a task-by-paradigm taxonomy. Its Sec. IV-C names the reference-versus-rollout split that Section I credits it with, and its Sec. III-F is the one bimanual subsection, 16 cited works with no coordination analysis.

The review of Kaup et al. [203] is the predecessor closest to the engine comparison, and it reviews Brax, Chrono, Gazebo, MuJoCo, ODE, PhysX, PyBullet, Unity and Webots for reinforcement learning research. It scores each on documentation, model and environment creation, URDF and MJCF support, and readiness for multi-agent work. It runs no benchmark of its own and says so in its Sec. V, that implementing the same scenarios across nine engines “goes beyond the scope of this paper”. Its running bodies are ant-and-humanoid RL benchmarks rather than hands, and it discusses no timestep, no friction model, no contact formulation and no penetration.

The oldest rows carry the cost of the corpus. Okamura et al. and Piazza et al. [204], [205], and Roa and Suárez [178], are behind publisher paywalls with no author-hosted copy found on 2026-09-18, and Bicchi [206] is in the same position with no row at all. Ma and Dollar [207] are a different case. The fetch that failed when its note was written succeeded afterwards, so a seven-page PDF is on disk with a recorded hash, and no note has been read from it. All five are cited by metadata only and nothing in this survey describes their contents. The open chapter [1] overlaps the paywalled Bicchi paper without being identical to it, so it is quoted in its own right.

A cell is a level and not a score: ✓ where the work gives the topic a structure of its own, a section, a table or a measurement it defines; ○ where the topic is there without one, a subsection with no analysis, a handful of mentions, or a metric named and never defined; × where it is absent or put outside the work’s scope. Each level coarsens one sentence read from `papers/notes/<key>.md`, and it is printed only while the quotation behind that sentence is still in the note, so a glyph cannot outlive its evidence. Appendix D prints the prose those levels coarsen; the cell-by-cell mapping is in `paper/tables/table10_surveys.md`. Three of the fourteen sources could not be obtained and one is on disk and has never been read, which is why their cells are empty; they are grouped as such rather than scored, because this survey knows nothing of them beyond a title and a DOI. The *gaps named* column is uniform by design: every predecessor names gaps, and what none of them attaches to a gap is a measurement.

APPENDIX E

WHERE THE PROTOCOL’S COUNTS COME FROM

Every count in Table III is derived below, and each axis is derived for the statistic that axis actually reports rather than by one convention applied to all of them: a single rate takes a Wilson half-width, a matched comparison takes McNemar, a ratio takes the standard error of the log ratio, and a correlation takes the Fisher-z interval. Section VII-C states what these derivations conclude. This survey re-ran no method, so every count here rests on an interval width, a power calculation or another paper’s measurement, and never on a measurement of our own.

Fix the width first, then read off the count. Take a 95 percent Wilson interval on a single reported rate, at the worst case of $\hat{p} = 0.5$. Writing the half-width of (1) as

$$w(n, \hat{p}) = \frac{z\sqrt{4n\hat{p}(1-\hat{p}) + z^2}}{2(n + z^2)}, \quad (2)$$

the count a target width w_0 demands is $n_{\min}(w_0) = \min\{n : w(n, 0.5) \leq w_0\}$, which gives

half-width w_0 (points)	20	15	10	5
$n_{\min}$ (trials)	21	39	93	381

Ten points is the coarsest width at which a single rate is worth printing, so the absolute-rate minimum is 93, rounded to 100. At 100 trials a reported 80 percent has an interval of 71 to 87 percent, and a reported 50 percent has 40 to 60. A comparison is a different question and a harder one: two rates each carrying ± 10 points do not resolve a 10-point difference between them, because the difference’s standard error is larger by a factor of $\sqrt{2}$, so the width argument sets a floor on what is worth reporting and not on what can be compared.

For the A/B comparison the relevant calculation is power, and the design is paired. Table III matches initial conditions by image overlay and interleaves the two policies in one session, so the unit is a matched pair and the count follows McNemar, which depends on the discordance rate. The share of initial conditions on which the two policies disagree, and not on the two rates alone. Write $\delta = |p_1 - p_2|$ for the gap between the two success rates, π_d for that discordance rate, $z_{1-\alpha/2} = 1.96$ at $\alpha = 0.05$ and $z_{1-\beta} = 0.84$ at 80 percent power; then the number of matched pairs is

$$n_{\text{pairs}} = \frac{[z_{1-\alpha/2}\sqrt{\pi_d} + z_{1-\beta}\sqrt{\pi_d - \delta^2}]^2}{\delta^2}. \quad (3)$$

To separate 50 from 70 percent, which is $\delta = 0.20$, at $\alpha = 0.05$ with 80 percent power, (3) asks per arm for

discordance π_d	0.2	0.3	0.4	0.5
n_{pairs}	37	57	77	96

The protocol assumes 0.3 and asks for 57, and states the sensitivity rather than hiding it, because 0.5 is the discordance the same two rates produce when the pairing buys nothing, and at that value the paired count returns to the 93 per arm an unpaired test would need. The saving from pairing is real but smaller than the pair counts suggest, since a pair costs two rollouts: 57 pairs is 114 rollouts against 186. An earlier version of this section quoted 93, 169 and 387 per arm for gaps of 20, 15 and 10 points, which are correct for independent arms and are the wrong test for this protocol. That 93 was also the same integer as the half-width calculation in the paragraph above, which is a coincidence of the worst-case arithmetic and not a second derivation of the same number.

One hundred is a cap and not a bill, because on a graded score a sequential test reached its decision in 12 to 36 paired hardware trials in [170]. At 30 rollouts a continuous score already carries a half-width of ± 0.36 standard deviations, which is why a graded score can stop where a binary one cannot. A cell that stops early does not report a Wilson interval. Optional stopping breaks the coverage of a fixed- n interval, which is the reason Snyder et al. [170] and Badithela et al. [173] use anytime-valid betting intervals rather than Wilson, so Table III asks a cell run to a fixed 100 for a Wilson interval and a cell stopped early for a confidence sequence, and never for both. Intervals are also

TABLE VII: What 14 predecessor surveys cover: ✓ covered, ○ partly, × not. 24 of 84 coverage cells are empty.

survey	yr	dexterous scope	taxonomy	bimanual	hardware	evaluation	gaps named
Surveys of the field							
Zhao et al. [175]	2026	✓	✓	○	✓	○	✓
An et al. [2]	2025	✓	○	○	○	○	✓
Bai et al. [37]	2025	○	✓	○	○	○	✓
Welte et al. [200]	2025	○	✓	○	✓	×	✓
Firoozi et al. [202]	2023	×	○	×	×	○	✓
Zhao et al. [201]	2020	×	✓	×	○	×	✓
Simulator and contact-model studies							
NVIDIA Isaac Sim [208]	2026	×	○	○	×	×	✓
Kaup et al. [203]	2024	×	✓	×	×	○	✓
Le Lidec et al. [35]	2023	×	✓	×	○	✓	✓
Erez et al. [90]	2015	×	×	×	○	✓	✓
Sources not obtained, or on disk and never read							
Piazza et al. [205]	2019	—	—	—	—	—	—
Roa and Suárez [178]	2015	—	—	—	—	—	—
Ma and Dollar [207]	2011	—	—	—	—	—	—
Okamura et al. [204]	2000	—	—	—	—	—	—

marginal rather than simultaneous. For example, evaluating twelve methods on seven axes produces 84 intervals. At 84 independent 95 percent intervals, four excursions are expected by construction, so a paper comparing k policies on m tasks corrects its $k(k-1)/2$ pairwise tests to a global 95 percent level, as the TRI LBM Team [179] does, or says its intervals are not simultaneous.

Simulation is cheap, so simulated cells take 200 episodes, giving a 6.9-point half-width. Perturbation axes are screened rather than certified, and 40 per axis buys a 15-point half-width on each axis’s own absolute rate, which is enough to rank the axes and pick the two worst for hardware. It is not enough for the ratio to the anchor that an earlier draft asked each cell to report. At 40 trials in each arm, a fall from a 0.50 anchor to 0.30 is a ratio of 0.60 with a 95 percent interval of 0.34 to 1.06, which contains 1: at the screening count you cannot establish that the perturbation hurt at all. Certifying that same drop takes 101 per arm at 80 percent power, or about 50 for an interval that merely excludes 1, so Table III now asks for absolute rates with their own intervals at 40, reports the ratio without an interval, and prescribes 101 before any claim that a named axis hurt.

For unseen objects the resampling unit is the object and not the trial, so 20 objects at 5 trials each gives 100 trials and an object-level half-width near 20 points. That 20 points is the Wilson width at $n = 20$ and it treats each object’s outcome as a single Bernoulli draw, which the five within-object trials are not. It is the right order of magnitude and the assumption belongs in the cell. A 10-point claim about an object distribution needs about 93 objects. Seven of the 39 rows that state an unseen count reach that: 225 in [42], 241 in [152] and [153], 360 in [11], 500 in [133], 2029 in [176] and 503409 in [48]. For a continuous score the half-width is 1.96 standard deviations over the square root of the count, $1.96 s/\sqrt{n}$, so 100 rollouts give ± 0.20 standard deviations, and the unit is the rollout because frames within one are correlated.

The transfer axis is the one where 100 is least defensible. On 100 matched pairs a measured correlation of 0.70 carries

a Fisher-z interval of 0.58 to 0.79, a half-width of about 0.10 that 130 pairs would be needed to guarantee. That is enough to establish that a simulator tracks reality at all, and it is not enough to separate [173]’s useful regime from its marginal one, since those differ by about 0.11 in correlation. Both limits come within 0.05 of the estimate only at about 457 pairs. Table III states which of the two decisions each count supports rather than leaving a reader to assume the larger one.

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